Riemanni hüpoteesist

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Part I
hüpotees

$\exists_{x_1}($
$\zeta(x_1)=0 \wedge$
$x_1 \in C \wedge$
$\neg(Re(x_1)=1/2) \wedge$
$\neg(\exists_{x_2}(x_2 \in N \wedge Re(x_1)=-x_2*2 \wedge Im(x_1)=0)$
$)$

Part II
muud seoud väited

1 zetafunktsioonist

1.1 kui reaalosa suurem kui 1

$\forall_{x_1}(x_1 \in C \wedge Re(x_1)<1 \rightarrow \zeta(-x_1)=\sum_{n=1}^{\infty}(n^{x_1})=\sum_{n=1}^{\infty}(n^{x_1})=\sum_{n=1}^{\infty}(n^{Re(x_1)+i*Im(x_1)})=\sum_{n=1}^{\infty}(n^{Re(x_1)}*n^{i*Im(x_1)})=$
$\sum_{n=1}^{\infty}(n^{Re(x_1)}*(cos(Im(x_1)*ln(n))+i*sin(Im(x_1)*ln(n)))))$

1.2 Dirichlet series

avaldis1: Dirichlet_series_Wikipediast

$\forall_{x_1}(x_1 \in C \wedge Re(x_1)>0 \rightarrow \zeta(x_1)=(x_1-1)^{-1}*\sum_{n=1}^{\infty}(\frac{n}{(n+1)^{x_1}}-\frac{n-x_1}{n^{x_1}})$

lihtsustan seda avaldist:

$\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_2 \in R \wedge x_1>0 \rightarrow$
$\zeta(x_1+i*x_2)=$

$(x_1-1+i*x_2)^{-1}*\sum_{n=1}^{\infty}(\frac{n}{(n+1)^{x_1+i*x_2}}-\frac{n-x_1-i*x_2}{n^{x_1+i*x_2}})=$
$(x_1-1+i*x_2)^{-1}*\sum_{n=1}^{\infty}(\frac{n}{(n+1)^{x_1}*(cos(x_2*ln(n+1))+i*sin(x_2*ln(n+1)))}-\frac{n-x_1-i*x_2}{n^{x_1}*(cos(x_2*ln(n))+i*sin(x_2*ln(n)))})=$

$\frac{x_1-1-i*x_2}{(x_1-1)^2+x_2^2}*\sum_{n=1}^{\infty}(n*(n+1)^{-x_1}*(cos(x_2*ln(n+1))-i*sin(x_2*ln(n+1)))-(n-x_1-i*x_2)*n^{-x_1}*(cos(x_2*$
$ln(n))-i*sin(x_2*ln(n))))=$

$\sum_{n=1}^{\infty}(((x_1-1)-i*x_2)*(cos(x_2*ln(n+1))-i*sin(x_2*ln(n+1)))*((x_1-1)^2+x_2^2)^{-1}*n*(n+1)^{-x_1}-$
$((x_1-1)-i*x_2)*(cos(x_2*ln(n))-i*sin(x_2*ln(n)))*((x_1-1)^2+x_2^2)^{-1}*(n-x_1-i*x_2)*n^{-x_1})=$

$\sum_{n=1}^{\infty}((((x_1-1)*cos(x_2*ln(n+1))-x_2*sin(x_2*ln(n+1)))-i*((x_1-1)*sin(x_2*ln(n+1))+x_2*cos(x_2*ln(n+1))))*((x_1-1)^2+x_2^2)^{-1}*n*(n+1)^{-x_1}-$
$((n-x_1)*((x_1-1)*cos(x_2*ln(n))-x_2*sin(x_2*ln(n)))-x_2*((x_1-1)*sin(x_2*ln(n))+x_2*cos(x_2*ln(n)))-i*$
$((n-x_1)*((x_1-1)*sin(x_2*ln(n))+x_2*cos(x_2*ln(n)))+x_2*(((x_1-1)*cos(x_2*ln(n))-x_2*sin(x_2*ln(n))))))*$
$((x_1-1)^2+x_2^2)^{-1}*n^{-x_1})=$

$\sum_{n=1}^{\infty}(((x_1-1)*cos(x_2*ln(n+1))-x_2*sin(x_2*ln(n+1)))*((x_1-1)^2+x_2^2)^{-1}*n*(n+1)^{-x_1}-((n-x_1)*((x_1-1)*cos(x_2*ln(n))-x_2*sin(x_2*ln(n)))-x_2*((x_1-1)*sin(x_2*ln(n))+x_2*cos(x_2*ln(n))))*((x_1-1)^2+x_2^2)^{-1}*n^{-x_1}$
$-i*(((x_1-1)*sin(x_2*ln(n+1))+x_2*cos(x_2*ln(n+1)))*((x_1-1)^2+x_2^2)^{-1}*n*(n+1)^{-x_1}+((n-x_1)*((x_1-1)*sin(x_2*ln(n))+x_2*cos(x_2*ln(n)))+x_2*(((x_1-1)*cos(x_2*ln(n))-x_2*sin(x_2*ln(n)))))*((x_1-1)^2+x_2^2)^{-1}*n^{-x_1}))=$
$((x_1-1)^2+x_2^2)^{-1}*\sum_{n=1}^{\infty}(((x_1-1)*cos(x_2*ln(n+1))-x_2*sin(x_2*ln(n+1)))*n*(n+1)^{-x_1}-((n-x_1)*((x_1-1)*cos(x_2*ln(n))-x_2*sin(x_2*ln(n)))-x_2*((x_1-1)*sin(x_2*ln(n))+x_2*cos(x_2*ln(n))))*n^{-x_1})-i*((x_1-1)^2+x_2^2)^{-1}*\sum_{n=1}^{\infty}((((x_1-1)*sin(x_2*ln(n+1))+x_2*cos(x_2*ln(n+1)))*n*(n+1)^{-x_1}-((n-x_1)*((x_1-1)*sin(x_2*ln(n))+x_2*cos(x_2*ln(n)))+x_2*((x_1-1)*cos(x_2*ln(n))-x_2*sin(x_2*ln(n))))*n^{-x_1}))=$
$))$

järgneva väite nimi on zeta2_reaalosa_ja_imaginaarosa_eraldatud:

$\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\rightarrow$
$\zeta(x_1+i*x_2)=$
$((x_1-1)^2+x_2^2)^{-1}*\sum_{n=1}^{\infty}($
$((x_1-1)*cos(x_2*ln(n+1))-x_2*sin(x_2*ln(n+1)))*n*(n+1)^{-x_1}-$
$((n-x_1)*((x_1-1)*cos(x_2*ln(n))-x_2*sin(x_2*ln(n)))-x_2*((x_1-1)*sin(x_2*ln(n))+x_2*cos(x_2*ln(n))))*n^{-x_1}$
$)$
$-i*((x_1-1)^2+x_2^2)^{-1}*\sum_{n=1}^{\infty}($
$((x_1-1)*sin(x_2*ln(n+1))+x_2*cos(x_2*ln(n+1)))*n*(n+1)^{-x_1}-$
$((n-x_1)*((x_1-1)*sin(x_2*ln(n))+x_2*cos(x_2*ln(n)))+x_2*((x_1-1)*cos(x_2*ln(n))-x_2*sin(x_2*ln(n))))*n^{-x_1}$
$)=$
$))$

lihtsustan seda väidet edasi

$\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\rightarrow$
$\zeta(x_1+i*x_2)=$

$(x_1^2-2*x_1+1+x_2^2)^{-1}*\sum_{n=1}^{\infty}($
$sin(x_2*ln(n)+0*(2\pi)/4)*x_2*(n-1)*n^{-x_1}-sin(x_2*ln(n+1)+0*(2\pi)/4)*x_2*n*(n+1)^{-x_1}$
$+sin(x_2*ln(n)+1*(2\pi)/4)*(x_1^2-x_1-n*x_1+n+x_2^2)*n^{-x_1}$
$-sin(x_2*ln(n+1)+1*(2\pi)/4)*(1-x_1)*n*(n+1)^{-x_1}$
$)$
$+i*(x_1^2-2*x_1+1+x_2^2)^{-1}*\sum_{n=1}^{\infty}($
$sin(x_2*ln(n)+1*(2\pi)/4)*x_2*(n-1)*n^{-x_1}-sin(x_2*ln(n+1)+1*(2\pi)/4)*x_2*n*(n+1)^{-x_1}$
$+sin(x_2*ln(n)+2*(2\pi)/4)*(x_1^2-x_1-n*x_1+n+x_2^2)*n^{-x_1}$
$-sin(x_2*ln(n+1)+2*(2\pi)/4)*(1-x_1)*n*(n+1)^{-x_1}$
$)$
$=$

$\lim_{M\rightarrow\infty}(\sum_{n=1}^{M}($
$+sin(x_2*ln(n)+1*(2\pi)/4)*(x_1^2-x_1-n*x_1+n+x_2^2)*n^{-x_1}$
$-sin(x_2*ln(n+1)+1*(2\pi)/4)*(1-x_1)*n*(n+1)^{-x_1}$
$)-sin(x_2*ln(M+1)+0*(2\pi)/4)*x_2*M*(M+1)^{-x_1})*$
$(x_1^2-2*x_1+1+x_2^2)^{-1}$
$+i*\lim_{M\rightarrow\infty}(\sum_{n=1}^{M}($
$sin(x_2*ln(n)+2*(2\pi)/4)*(x_1^2-x_1-n*x_1+n+x_2^2)*n^{-x_1}$
$-sin(x_2*ln(n+1)+2*(2\pi)/4)*(1-x_1)*n*(n+1)^{-x_1}$
$)-sin(x_2*ln(M+1)+1*(2\pi)/4)*x_2*M*(M+1)^{-x_1})*$
$(x_1^2-2*x_1+1+x_2^2)^{-1}=$

$\lim_{M\to\infty}(\sum_{n=1}^{M}($
$+sin(x_2*ln(n)+1*(2\pi)/4)*(x_1^2-2*x_1+x_2^2+1)*n^{-x_1}$
$+sin(x_2*ln(n)+1*(2\pi)/4)*(1-x_1)*(n-1)*n^{-x_1}-sin(x_2*ln(n+1)+1*(2\pi)/4)*(1-x_1)*n*(n+1)^{-x_1}$
$)-sin(x_2*ln(M+1)+0*(2\pi)/4)*x_2*M*(M+1)^{-x_1})*$
$(x_1^2-2*x_1+1+x_2^2)^{-1}$
$+i*\lim_{M\to\infty}(\sum_{n=1}^{M}($
$+sin(x_2*ln(n)+2*(2\pi)/4)*(x_1^2-2*x_1+x_2^2+1)*n^{-x_1}$
$+sin(x_2*ln(n)+2*(2\pi)/4)*(1-x_1)*(n-1)*n^{-x_1}-sin(x_2*ln(n+1)+2*(2\pi)/4)*(1-x_1)*n*(n+1)^{-x_1}$
$)-sin(x_2*ln(M+1)+1*(2\pi)/4)*x_2*M*(M+1)^{-x_1})*$
$(x_1^2-2*x_1+1+x_2^2)^{-1}=$
$\lim_{M\to\infty}($
$\sum_{n=1}^{M}(sin(x_2*ln(n)+1*(2\pi)/4)*n^{-x_1})$
$-(sin(x_2*ln(M+1)+1*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M+1)+0*(2\pi)/4)*x_2)*M*(M+1)^{-x_1}*((1-x_1)^2+$
$x_2^2)^{-1})$

$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{M}(sin(x_2*ln(n)+2*(2\pi)/4)*n^{-x_1})$
$-(sin(x_2*ln(M+1)+2*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M+1)+1*(2\pi)/4)*x_2)*M*(M+1)^{-x_1}*((1-x_1)^2+$
$x_2^2)^{-1})$
$=$

$\lim_{M\to\infty}($
$\sum_{n=1}^{M}(sin(x_2*ln(n)+1*(2\pi)/4)*n^{-x_1})$
$-(sin(x_2*ln(M+1)+1*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M+1)+0*(2\pi)/4)*x_2)*M*(M+1)^{-x_1}*(1+1/M)*$
$((1-x_1)^2+x_2^2)^{-1})$

$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{M}(sin(x_2*ln(n)+2*(2\pi)/4)*n^{-x_1})$
$-(sin(x_2*ln(M+1)+2*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M+1)+1*(2\pi)/4)*x_2)*M*(M+1)^{-x_1}*(1+1/M)*$
$((1-x_1)^2+x_2^2)^{-1})$
$=$

$\lim_{M\to\infty}($
$\sum_{n=1}^{M}(sin(x_2*ln(n)+1*(2\pi)/4)*n^{-x_1})$
$-(sin(x_2*ln(M+1)+1*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M+1)+0*(2\pi)/4)*x_2)*M^{1-x_1}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{M}(sin(x_2*ln(n)+2*(2\pi)/4)*n^{-x_1})$
$-(sin(x_2*ln(M+1)+2*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M+1)+1*(2\pi)/4)*x_2)*M^{1-x_1}*((1-x_1)^2+x_2^2)^{-1}$
$)=$

))

Valem zeta9_M_plus_1e_ei_ole:

$\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\to$
$\zeta(x_1+i*x_2)=$

$\lim_{M\to\infty}($
$\sum_{n=1}^{M}(sin(x_2*ln(n)+1*(2\pi)/4)*n^{-x_1})$
$-(sin(x_2*ln(M)+1*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M)+0*(2\pi)/4)*x_2)*M^{1-x_1}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{M}(sin(x_2*ln(n)+2*(2\pi)/4)*n^{-x_1})$
$-(sin(x_2*ln(M)+2*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M)+1*(2\pi)/4)*x_2)*M^{1-x_1}*((1-x_1)^2+x_2^2)^{-1}$
$)$

))

kuivõrd M läheneb lõpmatusele võin igal pool avaldises i asemele panna $floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})$ või floor(e$^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi/x_2}$).

$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow$
$\zeta(x_1 + i*x_2) =$

$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}(sin(x_2*ln(n)+1*(2\pi)/4)*n^{-x_1})$
$-(sin(x_2*ln(floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi}))+1*(2\pi)/4)*(1-x_1)+sin(x_2*ln(floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi}))+$
$0*(2\pi)/4)*x_2)*floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})^{1-x_1}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}(sin(x_2*ln(n)+2*(2\pi)/4)*n^{-x_1})$
$-(sin(x_2*ln(floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi}))+2*(2\pi)/4)*(1-x_1)+sin(x_2*ln(floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi}))+$
$1*(2\pi)/4)*x_2)*floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})^{1-x_1}*((1-x_1)^2+x_2^2)^{-1}$
$) =$

$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}(sin(x_2*ln(n)+1*(2\pi)/4)*n^{-x_1})$
$-(sin(x_2*ln((e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi}))+1*(2\pi)/4)*(1-x_1)+sin(x_2*ln((e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi}))+0*(2\pi)/4)*$
$x_2)*(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})^{1-x_1}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}(sin(x_2*ln(n)+2*(2\pi)/4)*n^{-x_1})$
$-(sin(x_2*ln((e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi}))+2*(2\pi)/4)*(1-x_1)+sin(x_2*ln((e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi}))+1*(2\pi)/4)*$
$x_2)*(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})^{1-x_1}*((1-x_1)^2+x_2^2)^{-1}$
$) =$

$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}(sin(x_2*ln(n)+1*(2\pi)/4)*n^{-x_1})$
$-(sin((floor(M)*(x_2/abs(x_2))+q_1)*2\pi+1*(2\pi)/4)*(1-x_1)+sin((floor(M)(x_2/abs(x_2))+q_1)*2\pi+0*$
$(2\pi)/4)*x_2)*(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)})*((1-x_1)^2+x_2^2)^{-1}$
$)$
$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}(sin(x_2*ln(n)+2*(2\pi)/4)*n^{-x_1})$
$-(sin((floor(M)*(x_2/abs(x_2))+q_2)*2\pi+2*(2\pi)/4)*(1-x_1)+sin((floor(M)*(x_2/abs(x_2))+q_2)*2\pi+1*$
$(2\pi)/4)*x_2)*(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)})*((1-x_1)^2+x_2^2)^{-1}$
$) =$
))))

zeta_funktsioon_2_q_ga:

$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow$
$\zeta(x_1 + i*x_2) =$
$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}(sin(x_2*ln(n)+(2\pi)*1/4)*n^{-x_1})$
$-(sin((q_1+1/4)*2\pi)*(1-x_1)+sin((q_1+0/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}(sin(x_2*ln(n)+(2\pi)*2/4)*n^{-x_1})$
$-(sin((q_2+2/4)*2\pi)*(1-x_1)+sin((q_2+1/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$) =$

))))

Nagu kvantrite abilgi öeldud on kehtib võrrand mistahes q_1 ja q_2 väärtuste puhul, aga kuna pole kindel, et millised väärtused q_1 ja q_2 asemele pannes saab kõige paremal kujul avaldise, siis loon su subsectionites võrrandist erinevaid variante, kus q_1 ja q_2'el on erinevad väärtused.

1.2.1 integraliga

kasutades seost $a*cos(t)+b*sin(t)=\sqrt{a^2+b^2}*sin(t+\arctan(\frac{a}{b}))$ lihtsustan eelnevat valemit:

$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\rightarrow$
$\zeta(x_1+i*x_2)=$
$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}(sin(x_2*ln(n)+(2\pi)*1/4)*n^{-x_1})$
$-sin(q_1*2\pi+\arctan((1-x_1)/x_2)*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1/2}$
$)$
$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}(sin(x_2*ln(n)+(2\pi)*2/4)*n^{-x_1})$
$-sin(q_2*2\pi-\arctan(x_2/(1-x_1)))*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1/2}$
$)$
))))

igalt summa elemendilt lahutan integraali eelmisest elemendist selle elemendine ja liidan maha lahutatud osa peale summerimist uuesti tagasi. Mingil põhjusel, mille chatgpt kirjutas tomib see ainult tingimusel $x_1<1$.

$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\wedge x_1<1\rightarrow$
$\zeta(x_1+i*x_2)=\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}(sin(x_2*ln(n)+(2\pi)*1/4)*n^{-x_1}-\int_{n-1}^{n}(dy*sin(x_2*ln(y)+(2\pi)*1/4)*y^{-x_1}))$
$+\int_0^{e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi}}(dy*sin(x_2*ln(y)+(2\pi)*1/4)*y^{-x_1})$
$+\int_{e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi}}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi}-1)}(dy*sin(x_2*ln(y)+(2\pi)*1/4)*y^{-x_1})$
$-sin(q_1*2\pi+\arctan((1-x_1)/x_2)*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1/2}$
$)$
$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}(sin(x_2*ln(n)+(2\pi)*2/4)*n^{-x_1}-\int_{n-1}^{n}(dy*sin(x_2*ln(y)+(2\pi)*2/4)*y^{-x_1}))$
$+\int_0^{e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi}}(dy*sin(x_2*ln(y)+(2\pi)*2/4)*y^{-x_1})$
$+\int_{e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi}}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-1}(dy*sin(x_2*ln(y)+(2\pi)*2/4)*y^{-x_1})$
$-sin(q_2*2\pi-\arctan(x_2/(1-x_1)))*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1/2}$
$)$
))))

asendan integrali sellega võrdse arihmeetilise avaldisega.

$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\wedge x_1<1\rightarrow$
$\zeta(x_1+i*x_2)=\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}($
$sin(x_2*ln(n)+(2\pi)*1/4)*n^{-x_1}$
$-(n^{1-x_1}*sin(x_2*ln(n)+arctan((1-x_1)/x_2))-(n-1)^{1-x_1}*sin(x_2*ln(n-1)+arctan((1-x_1)/x_2)))*((1-x_1)^2+x_2^2)^{-1/2}$
$)$
$+((e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})^{1-x_1}*sin(x_2*ln(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})+arctan((1-x_1)/x_2))-0^{1-x_1}*sin(x_2*ln(0)+arctan((1-x_1)/x_2)))*((1-x_1)^2+x_2^2)^{-1/2}$

$+((floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi}-1))^{1-x_1}*sin(x_2*ln(floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi}-1))+arctan((1-x_1)/x_2))-(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})^{1-x_1}*sin(x_2*ln(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})+arctan((1-x_1)/x_2)))*((1-x_1)^2+x_2^2)^{-1/2}$

$-sin(q_1*2\pi+\arctan((1-x_1)/x_2)*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1/2}$
)
$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}($
$sin(x_2*ln(n)+(2\pi)*2/4)*n^{-x_1}-$
$(n^{1-x_1}*sin(x_2*ln(n)-arctan(x_2/(1-x_1)))+(n-1)^{1-x_1}*sin(x_2*ln(n-1)-arctan(x_2/(1-x_1))))*((1-x_1)^2+x_2^2)^{-1/2}$
)
$+((e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})^{1-x_1}*sin(x_2*ln(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-arctan(x_2/(1-x_1)))+0^{1-x_1}*sin(x_2*ln(0)-arctan(x_2/(1-x_1))))*((1-x_1)^2+x_2^2)^{-1/2}$

$+((floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-1)^{1-x_1}*sin(x_2*ln(floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-1)-arctan(x_2/(1-x_1)))+(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})^{1-x_1}*sin(x_2*ln(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-arctan(x_2/(1-x_1))))*((1-x_1)^2+x_2^2)^{-1/2}$
$-sin(q_2*2\pi-\arctan(x_2/(1-x_1)))*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1/2}$
)
))))

lihtsustan. Kuna kui $x_1<1$ siis $\lim_{y\to0^+}(y^{1-x_1}*sin(x_2*ln(y)+arctan((1-x_1)/x_2)))=0$ ($0^{1-x_1}*sin(x_2*ln(0)+arctan((1-x_1)/x_2))=0$), sest $sin(x_2*ln(0)+arctan((1-x_1)/x_2))$ on määramatu, aga kindlalt piirides -1 kuni 1.

Kuna kui $x_1<1$ siis $\lim_{y\to0^+}(y^{1-x_1}*sin(x_2*ln(y)+arctan(x_2/(1-x_1))))=0$ ($0^{1-x_1}*sin(x_2*ln(0)-arctan(x_2/(1-x_1)))=0$),sest $sin(x_2*ln(0)-arctan(x_2/(1-x_1)))$ on määrmatu, aga kindlalt piirides -1 kuni 1.

SIIN ON VIGA!

$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\wedge x_1<1\to$
$\zeta(x_1+i*x_2)=\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}($
$sin(x_2*ln(n)+(2\pi)*1/4)*n^{-x_1}$
$-sin(x_2*ln(n)+arctan((1-x_1)/x_2))*n^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}$
$+sin(x_2*ln(n-1)+arctan((1-x_1)/x_2))*(n-1)^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}$
)
$+sin(q_1*2\pi+arctan((1-x_1)/x_2))*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1/2}$

$+((floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi}-1))^{1-x_1}*sin(x_2*ln(floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi}-1))+arctan((1-x_1)/x_2))-(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})^{1-x_1}*sin(x_2*ln(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})+arctan((1-x_1)/x_2)))*((1-x_1)^2+x_2^2)^{-1/2}$

$-sin(q_1*2\pi+\arctan((1-x_1)/x_2)*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1/2}$
)
$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}($
$sin(x_2*ln(n)+(2\pi)*2/4)*n^{-x_1}$
$-sin(x_2*ln(n)-arctan(x_2/(1-x_1)))*n^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}$
$+sin(x_2*ln(n-1)-arctan(x_2/(1-x_1)))*(n-1)^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}$
)
$+sin(q_2*2\pi-arctan(x_2/(1-x_1)))*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1/2}$

$+((floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-1)^{1-x_1}*sin(x_2*ln(floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-1)-arctan(x_2/(1-x_1)))+(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})^{1-x_1}*sin(q_2*2\pi-arctan(x_2/(1-x_1))))*((1-x_1)^2+x_2^2)^{-1/2}$
$-sin(q_2*2\pi-\arctan(x_2/(1-x_1)))*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1/2}$
)
))))

taandan

$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\wedge x_1<1\to$
$\zeta(x_1+i*x_2)=\lim_{M\to\infty}($

$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}($
$sin(x_2*ln(n)+(2\pi)*1/4)*n^{-x_1}$
$-sin(x_2*ln(n)+arctan((1-x_1)/x_2))*n^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}$
$+sin(x_2*ln(n-1)+arctan((1-x_1)/x_2))*(n-1)^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}$
$)$
$+($
$sin(x_2*ln(floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi}-1))+arctan((1-x_1)/x_2))*(floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi}-1))^{1-x_1}$
$-sin(x_2*ln(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})+arctan((1-x_1)/x_2))*(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})^{1-x_1}$
$)*((1-x_1)^2+x_2^2)^{-1/2}$

$)$
$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}($
$sin(x_2*ln(n)+(2\pi)*2/4)*n^{-x_1}$
$-sin(x_2*ln(n)-arctan(x_2/(1-x_1)))*n^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}$
$+sin(x_2*ln(n-1)-arctan(x_2/(1-x_1)))*(n-1)^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}$
$)$
$+($
$sin(x_2*ln(floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-1)-arctan(x_2/(1-x_1)))*(floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-1)^{1-x_1}$
$+sin(q_2*2\pi-arctan(x_2/(1-x_1)))*(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})^{1-x_1}$
$)*((1-x_1)^2+x_2^2)^{-1/2}$
$)$
$))))$

kuna kui M on väga suur, siis $\lim_{M\to 0}(ln(M-2)-ln(M)=0)\ \wedge\ \lim_{M\to 0}(M^x-(M-2)^x)=0$, siis

$sin(x_2*ln(floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi}-1))+arctan((1-x_1)/x_2))*(floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi}-1))^{1-x_1}$
$-sin(x_2*ln(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})+arctan((1-x_1)/x_2))*(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})^{1-x_1}=0$

ja

$sin(x_2*ln(floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-1)-arctan(x_2/(1-x_1)))*(floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-1)^{1-x_1}$
$+sin(q_2*2\pi-arctan(x_2/(1-x_1)))*(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})^{1-x_1}=0$

seega:

$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\wedge x_1<1\to\zeta(x_1+i*x_2)=$
$\lim_{M\to\infty}(\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}($
$sin(x_2*ln(n)+(2\pi)*1/4)*n^{-x_1}$
$-sin(x_2*ln(n)+arctan((1-x_1)/x_2))*n^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}$
$+sin(x_2*ln(n-1)+arctan((1-x_1)/x_2))*(n-1)^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}$
$))$
$+i*\lim_{M\to\infty}(\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}($
$sin(x_2*ln(n)+(2\pi)*2/4)*n^{-x_1}$
$-sin(x_2*ln(n)-arctan(x_2/(1-x_1)))*n^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}$
$+sin(x_2*ln(n-1)-arctan(x_2/(1-x_1)))*(n-1)^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}$
$))$
$))))$

kuna vastavate q_1 ja q_2 väärtuste valimise korral saab summa piiri suvaliselt nihutada, siis võib summa piiriks lihtsalt Mi panna

$\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\wedge x_1<1\to\zeta(x_1+i*x_2)=$
$\lim_{M\to\infty}(\sum_{n=1}^{M}($
$sin(x_2*ln(n)+(2\pi)*1/4)*n^{-x_1}$
$-sin(x_2*ln(n)+arctan((1-x_1)/x_2))*n^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}$
$+sin(x_2*ln(n-1)+arctan((1-x_1)/x_2))*(n-1)^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}$
$))$
$+i*\lim_{M\to\infty}(\sum_{n=1}^{M}($
$-sin(x_2*ln(n))*n^{-x_1}$

$$
\begin{aligned}
&-sin(x_2*ln(n)-arctan(x_2/(1-x_1)))*n^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}\\
&+sin(x_2*ln(n-1)-arctan(x_2/(1-x_1)))*(n-1)^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}\\
&))\\
&))
\end{aligned}
$$

Muudan summa piiri 1est 0iks:

$$
\begin{aligned}
&\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\wedge x_1<1\rightarrow \zeta(x_1+i*x_2)=\\
&\lim_{M\rightarrow\infty}(\sum_{n=0}^{M}(\\
&sin(x_2*ln(n+1)+(2\pi)*1/4)*(n+1)^{-x_1}\\
&-sin(x_2*ln(n+1)+arctan((1-x_1)/x_2))*(n+1)^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}\\
&+sin(x_2*ln(n)+arctan((1-x_1)/x_2))*n^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}\\
&))\\
&+i*\lim_{M\rightarrow\infty}(\sum_{n=0}^{M}(\\
&-sin(x_2*ln(n+1))*(n+1)^{-x_1}\\
&-sin(x_2*ln(n+1)-arctan(x_2/(1-x_1)))*(n+1)^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}\\
&+sin(x_2*ln(n)-arctan(x_2/(1-x_1)))*n^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}\\
&))\\
&))
\end{aligned}
$$

lihtsustan

$$
\begin{aligned}
&\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\wedge x_1<1\rightarrow \zeta(x_1+i*x_2)=\\
&\lim_{M\rightarrow\infty}(\sum_{n=0}^{M}(\\
&sin(x_2*ln(n+1)+(2\pi)*1/4)*(n+1)^{-x_1}\\
&-sin(x_2*ln(n+1)+arctan((1-x_1)/x_2))*(n+1)^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}\\
&+sin(x_2*ln(n)+arctan((1-x_1)/x_2))*n^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}\\
&))\\
&-i*\lim_{M\rightarrow\infty}(\sum_{n=0}^{M}(\\
&sin(x_2*ln(n+1))*(n+1)^{-x_1}\\
&-sin(x_2*ln(n+1)+arctan((1-x_1)/x_2))*(n+1)^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}\\
&+sin(x_2*ln(n)+arctan((1-x_1)/x_2))*n^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}\\
&))\\
&))
\end{aligned}
$$

1.2.2 edasine lihtsustamine lihtsamaid funktsioone sisaldama.

asendan siinuse selle Taylori reaga:

$$
\begin{aligned}
&\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\rightarrow\\
&\zeta(x_1+i*x_2)=\\
&\lim_{M\rightarrow\infty}(\\
&\sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}(\\
&{n_0}^{-x_1}*\sum_{n_1=0}^{\infty}((-1)^{n_1}*(x_2*ln(n_0))^{2*n_1}*factorial(2*n_1)^{-1})\\
&)\\
&-(sin((q_1+1/4)*2\pi)*(1-x_1)+sin((q_1+0/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}\\
&)\\
&+i*\lim_{M\rightarrow\infty}(\\
&-\sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}(\\
&{n_0}^{-x_1}*\sum_{n_1=0}^{\infty}((-1)^{n_1}*(x_2*ln(n_0))^{2*n_1+1}*factorial(2*n_1+1)^{-1})\\
&)\\
&-(sin((q_2+2/4)*2\pi)*(1-x_1)+sin((q_2+1/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}\\
&)\\
&))))
\end{aligned}
$$

muudan tegurite järjekorda:

$$
\begin{aligned}
&\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\rightarrow\\
&\zeta(x_1+i*x_2)=\\
&\lim_{M\rightarrow\infty}(
\end{aligned}
$$

$\sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}(n_0{}^{-x_1}*$
$\sum_{n_1=0}^{\infty}((-1)^{n_1}*x_2^{2*n_1}*factorial(2*n_1)^{-1}*ln(n_0)^{2*n_1})$
$)$
$-(sin((q_1+1/4)*2\pi)*(1-x_1)+sin((q_1+0/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$+i*\lim_{M\to\infty}($
$-\sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}(n_0{}^{-x_1}*$
$\sum_{n_1=0}^{\infty}((-1)^{n_1}*x_2^{2*n_1+1}*factorial(2*n_1+1)^{-1}*ln(n_0)^{2*n_1+1})$
$)$
$-(sin((q_2+2/4)*2\pi)*(1-x_1)+sin((q_2+1/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$))))$

asendal naturallogaritmi summaga:

$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\to$
$\zeta(x_1+i*x_2)=$
$\lim_{M\to\infty}($
$\sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}(n_0{}^{-x_1}*$
$\sum_{n_1=0}^{\infty}((-1)^{n_1}*x_2^{2*n_1}*factorial(2*n_1)^{-1}*(\sum_{n_2=0}^{\infty}((2*n_2+1)^{-1}*(n_0-1)^{2*n_2+1}*(n_0+1)^{-2*n_2-1}))^{2*n_1}*2^{2*n_1})$
$)$
$-(sin((q_1+1/4)*2\pi)*(1-x_1)+sin((q_1+0/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$+i*\lim_{M\to\infty}($
$-\sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}(n_0{}^{-x_1}*\sum_{n_1=0}^{\infty}((\sum_{n_2=0}^{\infty}((2*n_2+1)^{-1}*(n_0-1)^{2*n_2+1}*(n_0+1)^{-2*n_2-1}))^{2*n_1+1}*x_2^{2*n_1+1}*$
$factorial(2*n_1+1)^{-1}*2^{2*n_1+1}*(-1)^{n_1})$
$)$
$-(sin((q_2+2/4)*2\pi)*(1-x_1)+sin((q_2+1/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$))))$

kasutan multinomial valemit:

$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\wedge x_2\neq 0\to$
$\zeta(x_1+i*x_2)=$
$\lim_{M\to\infty}($
$\sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}(n_0{}^{-x_1}*\sum_{n_1=0}^{M^M-1}(x_2^{2*n_1}*2^{2*n_1}*(-1)^{n_1}*\sum_{n_2=0}^{(2*n_1+1)^{M^M}-1}($
$(\sum_{n_3=0}^{M^M-1}(\ floor(n_2/(2*n_1+1)^{n_3})\%(2*n_1+1)\ )=2*n_1)*$
$\Pi_{n_3=0}^{M^M-1}(factorial(\ floor(n_2/(2*n_1+1)^{n_3})\%(2*n_1+1)\ )^{-1}*((2*n_3+1)^{-1}*(n_0-1)^{2*n_3+1}*(n_0+1)^{-2*n_3-1})^{floor(n_2/(2*n_1+1)^{n_3})\%(2*n_1+1}$
$)))$
$-(sin((q_1+1/4)*2\pi)*(1-x_1)+sin((q_1+0/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$+i*\lim_{M\to\infty}($
$-\sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}(n_0{}^{-x_1}*\sum_{n_1=0}^{M^M-1}(x_2^{2*n_1+1}*2^{2*n_1+1}*(-1)^{n_1}*\sum_{n_2=0}^{(2*n_1+2)^{M^M}-1}($
$(\sum_{n_3=0}^{M^M-1}(\ floor(n_2/(2*n_1+2)^{n_3})\%(2*n_1+2)\ )=2*n_1+1)*$
$\Pi_{n_3=0}^{M^M-1}(factorial(\ floor(n_2/(2*n_1+2)^{n_3})\%(2*n_1+2)\ )^{-1}*((2*n_3+1)^{-1}*(n_0-1)^{2*n_3+1}*(n_0+1)^{-2*n_3-1})^{floor(n_2/(2*n_1+2)^{n_3})\%(2*n_1+2}$
$)))$
$-(sin((q_2+2/4)*2\pi)*(1-x_1)+sin((q_2+1/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$))))$

lihtsustan teades, et korutamise asemel võib astmed liita.

$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\wedge x_2\neq 0\to$
$\zeta(x_1+i*x_2)=$

$\lim_{M\to\infty}($
$\sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}(n_0{}^{-x_1}*\sum_{n_1=0}^{M^M-1}(x_2^{2*n_1}*2^{2*n_1}*(-1)^{n_1}*\sum_{n_2=0}^{(2*n_1+1)^{M^M}-1}($
$(\sum_{n_3=0}^{M^M-1}(\ floor(n_2/(2*n_1+1)^{n_3})\%(2*n_1+1)\ )=2*n_1)*$
$(n_0-1)^{\sum_{n_3=0}^{M^M-1}((2*n_3+1)*(floor(n_2/(2*n_1+1)^{n_3})\%(2*n_1+1)))}*$
$(n_0+1)^{-\sum_{n_3=0}^{M^M-1}((2*n_3+1)*(floor(n_2/(2*n_1+1)^{n_3})\%(2*n_1+1)))}*$
$\Pi_{n_3=0}^{M^M-1}(factorial(\ floor(n_2/(2*n_1+1)^{n_3})\%(2*n_1+1)\ )^{-1}*(2*n_3+1)^{-(floor(n_2/(2*n_1+1)^{n_3})\%(2*n_1+1))})$
$)))$
$-(sin((q_1+1/4)*2\pi)*(1-x_1)+sin((q_1+0/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$+i*\lim_{M\to\infty}($
$-\sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}(n_0{}^{-x_1}*\sum_{n_1=0}^{M^M-1}(x_2^{2*n_1+1}*2^{2*n_1+1}*(-1)^{n_1}*\sum_{n_2=0}^{(2*n_1+2)^{M^M}-1}($
$(\sum_{n_3=0}^{M^M-1}(\ floor(n_2/(2*n_1+2)^{n_3})\%(2*n_1+2)\ )=2*n_1+1)*$
$(n_0-1)^{\sum_{n_3=0}^{M^M-1}((2*n_3+1)*(floor(n_2/(2*n_1+2)^{n_3})\%(2*n_1+2)))}*$
$(n_0+1)^{-\sum_{n_3=0}^{M^M-1}((2*n_3+1)*(floor(n_2/(2*n_1+2)^{n_3})\%(2*n_1+2)))}*$
$\Pi_{n_3=0}^{M^M-1}(factorial(\ floor(n_2/(2*n_1+2)^{n_3})\%(2*n_1+2)\ )^{-1}*(2*n_3+1)^{-(floor(n_2/(2*n_1+2)^{n_3})\%(2*n_1+2))})$
$)))$
$-(sin((q_2+2/4)*2\pi)*(1-x_1)+sin((q_2+1/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$))))$

selle asemel et üle erinevate n2'ede ristsummade summeeridaarvutan lihtsalt n2'e ristsumma:

$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\wedge x_2\neq 0\rightarrow$
$\zeta(x_1+i*x_2)=$
$\lim_{M\to\infty}($
$\sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}(n_0^{-x_1}*\sum_{n_1=0}^{(2*M^M)^{M^M}-1}($
$((1+\sum_{n_2=0}^{M^M-1}(floor(n_1/(2*M^M)^{n_2})\%(2*M^M)))\%2)*$
$(\sum_{n_2=0}^{M^M-1}(floor(n_1/(2*M^M)^{n_2})\%(2*M^M))<2*M^M-1)*$
$(x_2*2)^{\sum_{n_2=0}^{M^M-1}(floor(n_1/(2*M^M)^{n_2})\%(2*M^M))}*$
$(-1)^{\sum_{n_2=0}^{M^M-1}(floor(n_1/(2*M^M)^{n_2})\%(2*M^M))/2}*$
$(n_0-1)^{\sum_{n_2=0}^{M^M-1}((2*n_2+1)*(floor(n_1/(2*M^M)^{n_2})\%(2*M^M)))}*$
$(n_0+1)^{-\sum_{n_2=0}^{M^M-1}((2*n_2+1)*(floor(n_1/(2*M^M)^{n_2})\%(2*M^M)))}*$
$\Pi_{n_2=0}^{M^M-1}(factorial(floor(n_1/(2*M^M)^{n_2})\%(2*M^M))^{-1}*(2*n_2+1)^{-(floor(n_1/(2*M^M)^{n_2})\%(2*M^M))})$
$))$
$-(sin((q_1+1/4)*2\pi)*(1-x_1)+sin((q_1+0/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$+i*\lim_{M\to\infty}($
$\sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}(n_0^{-x_1}*\sum_{n_1=0}^{(2*M^M)^{M^M}-1}($
$(\sum_{n_2=0}^{M^M-1}(floor(n_1/(2*M^M)^{n_2})\%(2*M^M))\%2)*$
$(\sum_{n_2=0}^{M^M-1}(floor(n_1/(2*M^M)^{n_2})\%(2*M^M))<2*M^M)*$
$(x_2*2)^{\sum_{n_2=0}^{M^M-1}(floor(n_1/(2*M^M)^{n_2})\%(2*M^M))}*$
$(-1)^{(\sum_{n_2=0}^{M^M-1}(floor(n_1/(2*M^M)^{n_2})\%(2*M^M))+3)/2+1}*$
$(n_0-1)^{\sum_{n_2=0}^{M^M-1}((2*n_2+1)*(floor(n_1/(2*M^M)^{n_2})\%(2*M^M)))}*$
$(n_0+1)^{-\sum_{n_2=0}^{M^M-1}((2*n_2+1)*(floor(n_1/(2*M^M)^{n_2})\%(2*M^M)))}*$

$\Pi_{n_2=0}^{M^M-1}(factorial(floor(n_1/(2*M^M)^{n_2})\%(2*M^M))^{-1}*(2*n_2+1)^{-(floor(n_1/(2*M^M)^{n_2})\%(2*M^M))})$
$))$
$-(sin((q_2+2/4)*2\pi)*(1-x_1)+sin((q_2+1/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$))))$

kuna Mi lähenedes lõpmatusele see avaldis nagu nii koondub ja selle muutuse tegemine ainult lisab liikmeid, mis on absoluutväärtuselt väiksemad kui oleksid liikmed, mis lisanduksid Mi suurendamisel:

$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\wedge x_2\neq 0\rightarrow$
$\zeta(x_1+i*x_2)=$
$\lim_{M\rightarrow\infty}($
$\sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}(n_0^{-x_1}*\sum_{n_1=0}^{(2*M^M)^{M^M}-1}($
$((1+\sum_{n_2=0}^{M^M-1}(floor(n_1/(2*M^M)^{n_2})\%(2*M^M)))\%2)*$
$(x_2*2)^{\sum_{n_2=0}^{M^M-1}(floor(n_1/(2*M^M)^{n_2})\%(2*M^M))}*$
$(-1)^{\sum_{n_2=0}^{M^M-1}(floor(n_1/(2*M^M)^{n_2})\%(2*M^M))/2}*$
$(n_0-1)^{\sum_{n_2=0}^{M^M-1}((2*n_2+1)*(floor(n_1/(2*M^M)^{n_2})\%(2*M^M)))}*$
$(n_0+1)^{-\sum_{n_2=0}^{M^M-1}((2*n_2+1)*(floor(n_1/(2*M^M)^{n_2})\%(2*M^M)))}*$
$\Pi_{n_2=0}^{M^M-1}(factorial(floor(n_1/(2*M^M)^{n_2})\%(2*M^M))^{-1}*(2*n_2+1)^{-(floor(n_1/(2*M^M)^{n_2})\%(2*M^M))})$
$))$
$-(sin((q_1+1/4)*2\pi)*(1-x_1)+sin((q_1+0/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$+i*\lim_{M\rightarrow\infty}($
$\sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}(n_0^{-x_1}*\sum_{n_1=0}^{(2*M^M)^{M^M}-1}($
$(\sum_{n_2=0}^{M^M-1}(floor(n_1/(2*M^M)^{n_2})\%(2*M^M))\%2)*$
$(x_2*2)^{\sum_{n_2=0}^{M^M-1}(floor(n_1/(2*M^M)^{n_2})\%(2*M^M))}*$
$(-1)^{(\sum_{n_2=0}^{M^M-1}(floor(n_1/(2*M^M)^{n_2})\%(2*M^M))+3)/2+1}*$
$(n_0-1)^{\sum_{n_2=0}^{M^M-1}((2*n_2+1)*(floor(n_1/(2*M^M)^{n_2})\%(2*M^M)))}*$
$(n_0+1)^{-\sum_{n_2=0}^{M^M-1}((2*n_2+1)*(floor(n_1/(2*M^M)^{n_2})\%(2*M^M)))}*$
$\Pi_{n_2=0}^{M^M-1}(factorial(floor(n_1/(2*M^M)^{n_2})\%(2*M^M))^{-1}*(2*n_2+1)^{-(floor(n_1/(2*M^M)^{n_2})\%(2*M^M))})$
$))$
$-(sin((q_2+2/4)*2\pi)*(1-x_1)+sin((q_2+1/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$))))$

viin $n_0^{-x_1}$ summa sisse ja panen n_0 summa 0 ist algama.

$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\wedge x_2\neq 0\rightarrow$
$\zeta(x_1+i*x_2)=$
$\lim_{M\rightarrow\infty}($
$\sum_{n_0=0}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})-1}(\sum_{n_1=0}^{(2*M^M)^{M^M}-1}($
$((1+\sum_{n_2=0}^{M^M-1}(floor(n_1/(2*M^M)^{n_2})\%(2*M^M)))\%2)*$
$(x_2*2)^{\sum_{n_2=0}^{M^M-1}(floor(n_1/(2*M^M)^{n_2})\%(2*M^M))}*$
$(-1)^{\sum_{n_2=0}^{M^M-1}(floor(n_1/(2*M^M)^{n_2})\%(2*M^M))/2}*$
$n_0^{\sum_{n_2=0}^{M^M-1}((2*n_2+1)*(floor(n_1/(2*M^M)^{n_2})\%(2*M^M)))}*$
$(n_0+1)^{-x_1}*$
$(n_0+2)^{-\sum_{n_2=0}^{M^M-1}((2*n_2+1)*(floor(n_1/(2*M^M)^{n_2})\%(2*M^M)))}*$
$\Pi_{n_2=0}^{M^M-1}(factorial(floor(n_1/(2*M^M)^{n_2})\%(2*M^M))^{-1}*(2*n_2+1)^{-(floor(n_1/(2*M^M)^{n_2})\%(2*M^M))})$

$$
))
-(sin((q_1+1/4)*2\pi)*(1-x_1)+sin((q_1+0/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}
)
+i*\lim_{M\to\infty}(
\sum_{n_0=0}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-1}(\sum_{n_1=0}^{(2*M^M)^{M^M}-1}(
(\sum_{n_2=0}^{M^M-1}(floor(n_1/(2*M^M)^{n_2})\%(2*M^M))\%2)*
(x_2*2)^{\sum_{n_2=0}^{M^M-1}(floor(n_1/(2*M^M)^{n_2})\%(2*M^M))}*
(-1)^{(\sum_{n_2=0}^{M^M-1}(floor(n_1/(2*M^M)^{n_2})\%(2*M^M))+3)/2+1}*
n_0^{\sum_{n_2=0}^{M^M-1}((2*n_2+1)*(floor(n_1/(2*M^M)^{n_2})\%(2*M^M)))}*
(n_0+1)^{-x_1}*
(n_0+2)^{-\sum_{n_2=0}^{M^M-1}((2*n_2+1)*(floor(n_1/(2*M^M)^{n_2})\%(2*M^M)))}*
\Pi_{n_2=0}^{M^M-1}(factorial(floor(n_1/(2*M^M)^{n_2})\%(2*M^M))^{-1}*(2*n_2+1)^{-(floor(n_1/(2*M^M)^{n_2})\%(2*M^M))})
))
-(sin((q_2+2/4)*2\pi)*(1-x_1)+sin((q_2+1/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}
)
))))
$$

lihtsustan viies n0-summa n1-summa sisse. ja muudan summade indekseid.

$$
\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\wedge x_2\neq 0\to
\zeta(x_1+i*x_2)=
\lim_{M\to\infty}(\sum_{n_0=0}^{(2*M^M)^{M^M}-1}(
((1+\sum_{n_1=0}^{M^M-1}(floor(n_0/(2*M^M)^{n_1})\%(2*M^M)))\%2)*
(x_2*2)^{\sum_{n_1=0}^{M^M-1}(floor(n_0/(2*M^M)^{n_1})\%(2*M^M))}*
(-1)^{\sum_{n_1=0}^{M^M-1}(floor(n_0/(2*M^M)^{n_1})\%(2*M^M))/2}*
\sum_{n_1=0}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})-1}(
n_1^{\sum_{n_2=0}^{M^M-1}((2*n_2+1)*(floor(n_0/(2*M^M)^{n_2})\%(2*M^M)))}*
(n_1+1)^{-x_1}*
(n_1+2)^{-\sum_{n_2=0}^{M^M-1}((2*n_2+1)*(floor(n_0/(2*M^M)^{n_2})\%(2*M^M)))}
)*
\Pi_{n_1=0}^{M^M-1}(factorial(floor(n_0/(2*M^M)^{n_1})\%(2*M^M))^{-1}*(2*n_1+1)^{-(floor(n_0/(2*M^M)^{n_1})\%(2*M^M))})
)
-(sin((q_1+1/4)*2\pi)*(1-x_1)+sin((q_1+0/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}
)
+i*\lim_{M\to\infty}(
\sum_{n_0=0}^{(2*M^M)^{M^M}-1}(
(\sum_{n_1=0}^{M^M-1}(floor(n_0/(2*M^M)^{n_1})\%(2*M^M))\%2)*
(x_2*2)^{\sum_{n_1=0}^{M^M-1}(floor(n_0/(2*M^M)^{n_1})\%(2*M^M))}*
(-1)^{(\sum_{n_1=0}^{M^M-1}(floor(n_0/(2*M^M)^{n_1})\%(2*M^M))+3)/2+1}*
\sum_{n_1=0}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-1}(
n_1^{\sum_{n_2=0}^{M^M-1}((2*n_2+1)*(floor(n_0/(2*M^M)^{n_2})\%(2*M^M)))}*
(n_1+1)^{-x_1}*
(n_1+2)^{-\sum_{n_2=0}^{M^M-1}((2*n_2+1)*(floor(n_0/(2*M^M)^{n_2})\%(2*M^M)))}
)*
$$

$\Pi_{n_1=0}^{M^M-1}(factorial(floor(n_0/(2*M^M)^{n_1})\%(2*M^M))^{-1}*(2*n_1+1)^{-(floor(n_0/(2*M^M)^{n_1})\%(2*M^M))})$
$)$
$-(sin((q_2+2/4)*2\pi)*(1-x_1)+sin((q_2+1/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$))))$

lihtsustan n1-summa(vana n0-summa) sees olevat osa:

$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\wedge x_2\neq 0\rightarrow$
$\zeta(x_1+i*x_2)=$
$\lim_{M\to\infty}(\sum_{n_0=0}^{(2*M^M)^{M^M}-1}($
$((1+\sum_{n_1=0}^{M^M-1}(floor(n_0/(2*M^M)^{n_1})\%(2*M^M)))\%2)*$
$(x_2*2)^{\sum_{n_1=0}^{M^M-1}(floor(n_0/(2*M^M)^{n_1})\%(2*M^M))}*$
$(-1)^{\sum_{n_1=0}^{M^M-1}(floor(n_0/(2*M^M)^{n_1})\%(2*M^M))/2}*$
$\sum_{n_1=2}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})+1}($
$(1-2/n_1)^{\sum_{n_2=0}^{M^M-1}((2*n_2+1)*(floor(n_0/(2*M^M)^{n_2})\%(2*M^M)))}*$
$(n_1-1)^{-x_1}$
$)*$
$\Pi_{n_1=0}^{M^M-1}(factorial(floor(n_0/(2*M^M)^{n_1})\%(2*M^M))^{-1}*(2*n_1+1)^{-(floor(n_0/(2*M^M)^{n_1})\%(2*M^M))})$
$)$
$-(sin((q_1+1/4)*2\pi)*(1-x_1)+sin((q_1+0/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$+i*\lim_{M\to\infty}($
$\sum_{n_0=0}^{(2*M^M)^{M^M}-1}($
$(\sum_{n_1=0}^{M^M-1}(floor(n_0/(2*M^M)^{n_1})\%(2*M^M))\%2)*$
$(x_2*2)^{\sum_{n_1=0}^{M^M-1}(floor(n_0/(2*M^M)^{n_1})\%(2*M^M))}*$
$(-1)^{(\sum_{n_1=0}^{M^M-1}(floor(n_0/(2*M^M)^{n_1})\%(2*M^M))+3)/2+1}*$
$\sum_{n_1=2}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})+1}($
$(1-2/n_1)^{\sum_{n_2=0}^{M^M-1}((2*n_2+1)*(floor(n_0/(2*M^M)^{n_2})\%(2*M^M)))}*$
$(n_1-1)^{-x_1}$
$)*$
$\Pi_{n_1=0}^{M^M-1}(factorial(floor(n_0/(2*M^M)^{n_1})\%(2*M^M))^{-1}*(2*n_1+1)^{-(floor(n_0/(2*M^M)^{n_1})\%(2*M^M))})$
$)$
$-(sin((q_2+2/4)*2\pi)*(1-x_1)+sin((q_2+1/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$))))$

1.2.3 tagumine pool nulliks

$sin((q_1+1/4)*2\pi)*(1-x_1)+sin((q_1+0/4)*2\pi)*x_2=0\wedge$
$sin((q_2+2/4)*2\pi)*(1-x_1)+sin((q_2+1/4)*2\pi)*x_2=0$

$\sin((q_1+1/4)*2\pi)*(1-x_1)+sin(q_1*2\pi)*x_2=0\wedge$
$sin((q_2+2/4)*2\pi)*(1-x_1)+sin((q_2+1/4)*2\pi)*x_2=0$

$\exists_k(k\in N\wedge q_1=k/2-arctan((1-x_1)/x_2)/(2\pi))$
$\exists_k(k\in N\wedge q_2=k/2+arctan(x_2/(1-x_1))/(2\pi))$

$\exists_k(k \in N \wedge q_2 = k/2 - arctan((1-x_1)/x_2)/(2\pi) + 1/4)$

$arctan(x_2/(1-x_1)) + arctan((1-x_1)/x_2) = 2\pi/4 * sgn((1-x_1)*x_2)$
$arctan(x_2/(1-x_1)) = -arctan((1-x_1)/x_2) + 2\pi/4 * sgn((1-x_1)*x_2)$

$\exists_k(k \in N \wedge q_2 = k/2 - arctan((1-x_1)/x_2)/(2\pi) + 1/4 * sgn((1-x_1)*x_2))$

$q_2 = q_1 + 1/4 * sgn((1-x_1)*x_2)) + k/2$
$q_2 = q_1 + 1/4$
$\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow$
$\zeta(x_1 + i*x_2) =$

$\lim_{M\to\infty}(\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+(k/2-arctan((1-x_1)/x_2)/(2\pi))/x_2)*2\pi})}(sin((x_2*ln(n)/(2\pi)+1/4)*2\pi)*n^{-x_1}))$
$+ i * \lim_{M\to\infty}(-\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+(k/2+arctan(x_2/(1-x_1))/(2\pi))/x_2)*2\pi})}(sin((x_2*ln(n)/(2\pi))*2\pi)*n^{-x_1})) =$

$\lim_{M\to\infty}(\sum_{n=1}^{floor(e^{(floor(M)*2\pi/abs(x_2)+k*2\pi/2-arctan((1-x_1)/x_2)/x_2)})}(sin(x_2*ln(n)+2\pi/4)*n^{-x_1}))$
$+ i * \lim_{M\to\infty}(-\sum_{n=1}^{floor(e^{(floor(M)*2\pi/abs(x_2)+k*2\pi/2+arctan(x_2/(1-x_1))/x_2)})}(sin(x_2*ln(n))*n^{-x_1})) =$

$\lim_{M\to\infty}(\sum_{n=1}^{floor(e^{(floor(M)*2\pi/2/abs(x_2)-arctan((1-x_1)/x_2)/x_2)})}(sin(x_2*ln(n)+2\pi/4)*n^{-x_1}))$
$+ i * \lim_{M\to\infty}(-\sum_{n=1}^{floor(e^{(floor(M)*2\pi/2/abs(x_2)+arctan(x_2/(1-x_1))/x_2)})}(sin(x_2*ln(n))*n^{-x_1})) =$

$\lim_{M\to\infty}(\sum_{n=1}^{floor(e^{(floor(M)*2\pi/2/abs(x_2)-arctan((1-x_1)/x_2)/x_2)})}(sin(x_2*ln(n)+2\pi/4)*n^{-x_1}))$
$+ i * \lim_{M\to\infty}(-\sum_{n=1}^{floor(e^{(floor(M)*2\pi/2/abs(x_2)+(-arctan((1-x_1)/x_2)/x_2+2\pi/4*sgn((1-x_1)*x_2))/x_2)})}(sin(x_2*ln(n))*n^{-x_1})) =$

$\lim_{M\to\infty}(\sum_{n=1}^{floor(e^{floor(M)*2\pi/2/abs(x_2)-arctan((1-x_1)/x_2)/x_2)})}(sin(x_2*ln(n)+2\pi/4)*n^{-x_1}))$
$+ i * \lim_{M\to\infty}(-\sum_{n=1}^{floor(e^{floor(M)*2\pi/2/abs(x_2)-arctan((1-x_1)/x_2)/x_2+2\pi/4*sgn((1-x_1)*x_2)/x_2)})}(sin(x_2*ln(n))*n^{-x_1})) =$

$))$

Valem zeta13_tagumine_pool_nulliks:

$\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow$
$\zeta(x_1 + i*x_2) =$
$\lim_{M\to\infty}(\sum_{n=1}^{floor(e^{floor(M)/2*2\pi/abs(x_2)-arctan((1-x_1)/x_2)/x_2)})}(sin(x_2*ln(n)+2\pi/4)*n^{-x_1}))$
$+ i * \lim_{M\to\infty}(-\sum_{n=1}^{floor(e^{(floor(M)/2+1/4)*2\pi/abs(x_2)-arctan((1-x_1)/x_2)/x_2)})}(sin(x_2*ln(n))*n^{-x_1}))$
$))$

asendan siinuse ja logaritmi vastavate Taylori ridadega.

$\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow$
$\zeta(x_1 + i*x_2) =$
$\lim_{M\to\infty}(\sum_{n_0=1}^{floor(e^{floor(M)/2*2\pi/abs(x_2)-arctan((1-x_1)/x_2)/x_2)}}($
$\sum_{n_1=0}^{M}($
$(-1)^{n_1} * (fact(2*n_1))^{-1} * (x_2*ln(n_0)+2\pi/4)^{2*n_1} * n_0^{-x_1})))$
$+ i * \lim_{M\to\infty}(-\sum_{n_0=1}^{floor(e^{(floor(M)/2+1/4)*2\pi/abs(x_2)-arctan((1-x_1)/x_2)/x_2)})}(sin(x_2*ln(n_0))*n_0^{-x_1}))$
$))$

asendan logaritmi vastavate Taylori ridadega.

valem zeta14_tagumine_pool_nulliks_tayloriga(ei suutnud kontrollida, aga vb sest arvutusvõime piiratuse tõttu ei saanud piidsavaölt suurt Mi vsalida):

$\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow$
$\zeta(x_1 + i*x_2) =$
$\lim_{M\to\infty}(\sum_{n_0=0}^{floor(e^{floor(M)/2*2\pi/abs(x_2)-arctan((1-x_1)/x_2)/x_2)})}(\sum_{n_1=0}^{(2*M)^M-1}($
$(1+\%(2*M))\%2)*$

$(x_2*2)^{\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))}*$
$(-1)^{\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))/2}*$
$n_0^{\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_1/(2*M)^{n_2})\%(2*M)))}*$
$(n_0+1)^{-x_1}*$
$(n_0+2)^{-\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_1/(2*M)^{n_2})\%(2*M)))}*$
$\Pi_{n_2=0}^{M-1}(factorial(floor(n_1/(2*M)^{n_2})\%(2*M))^{-1}*(2*n_2+1)^{-floor(n_1/(2*M)^{n_2})\%(2*M)})$
)))

+i*$\lim_{M\to\infty}(\sum_{n_0=0}^{floor(e^{(floor(M)/2+1/4)*2\pi/abs(x_2)-arctan((1-x_1)/x_2)/x_2})}(\sum_{n_1=0}^{(2*M)^M-1}($
$\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))\%2)*$
$(x_2*2)^{\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))}*$
$(-1)^{(\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))+1)/2}*$
$n_0^{\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_1/(2*M)^{n_2})\%(2*M)))}*$
$(n_0+1)^{-x_1}*$
$(n_0+2)^{-\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_1/(2*M)^{n_2})\%(2*M)))}*$
$\Pi_{n_2=0}^{M-1}(factorial(floor(n_1/(2*M)^{n_2})\%(2*M))^{-1}*(2*n_2+1)^{-floor(n_1/(2*M)^{n_2})\%(2*M)})$
)))

))

1.2.4 $q_1 = q_2$

$\forall_{q_1}(\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \to$

$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q/x_2)*2\pi})}(sin(x_2*ln(n)+(2\pi)*1/4)*n^{-x_1})$
$-(sin((q+1/4)*2\pi)*(1-x_1)+sin((q+0/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q/x_2)*2\pi*(1-x_1)/x_2}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q/x_2)*2\pi})}(sin(x_2*ln(n)+(2\pi)*2/4)*n^{-x_1})$
$-(sin((q+2/4)*2\pi)*(1-x_1)+sin((q+1/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q/x_2)*2\pi*(1-x_1)/x_2}*((1-x_1)^2+x_2^2)^{-1}$
$)$
))))

$q_1 = 0 \wedge q_2 = 0$ kui valida, et $q_1 = 0$ ja $q_2 = 0$. Valem zeta12:
$\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \to$
$\zeta(x_1 + i*x_2) =$

$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{floor(M)*2\pi/abs(x_2)})}(sin(x_2*ln(n)+1*(2\pi)/4)*n^{-x_1})$
$-e^{floor(M)*2\pi*(1-x_1)/abs(x_2)}*((1-x_1)^2+x_2^2)^{-1}*(1-x_1)$
$)$
$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{floor(M)*2\pi/abs(x_2)})}(sin(x_2*ln(n)+2*(2\pi)/4)*n^{-x_1})$
$-e^{floor(M)*2\pi*(1-x_1)/abs(x_2)}*((1-x_1)^2+x_2^2)^{-1}*x_2$
$)=$
))

1.2.5 $q_1 = 1/8 \wedge q_2 = -1/8$

$\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow$
$\zeta(x_1 + i * x_2) =$

$$\lim_{M\to\infty}(
\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+1/8/x_2)*2\pi})}(sin((x_2 * ln(n)/(2\pi) + 1/4) * 2\pi) * n^{-x_1})
- (sin((1/8+1/4)*2\pi)*(1-x_1) + sin((1/8+0/4)*2\pi)*x_2) * e^{(floor(M)/abs(x_2)+1/8/x_2)*2\pi*(1-x_1)} * ((1-x_1)^2 + x_2^2)^{-1}
)
+ i * \lim_{M\to\infty}(
\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)-1/8/x_2)*2\pi})}(sin((x_2 * ln(n)/(2\pi) + 2/4) * 2\pi) * n^{-x_1})
- (sin((-1/8+2/4)*2\pi)*(1-x_1) + sin((-1/8+1/4)*2\pi)*x_2) * e^{(floor(M)/abs(x_2)-1/8/x_2)*2\pi*(1-x_1)} * ((1-x_1)^2 + x_2^2)^{-1}
) =$$

$$\lim_{M\to\infty}(
\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+1/8/x_2)*2\pi})}(sin((x_2 * ln(n)/(2\pi) + 1/4) * 2\pi) * n^{-x_1})
- ((1-x_1) + x_2) * e^{(floor(M)/abs(x_2)+1/8/x_2)*2\pi*(1-x_1)} * ((1-x_1)^2 + x_2^2)^{-1} * 2^{-1/2}
)
+ i * \lim_{M\to\infty}(
\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)-1/8/x_2)*2\pi})}(sin((x_2 * ln(n)/(2\pi) + 2/4) * 2\pi) * n^{-x_1})
- ((1-x_1) + x_2) * e^{(floor(M)/abs(x_2)-1/8/x_2)*2\pi*(1-x_1)} * ((1-x_1)^2 + x_2^2)^{-1} * 2^{-1/2}
)$$

))

1.2.6 kirjutades siinuse ja koosiinuse summana

kasutatud valem $sin(x) = \sum_{m=0}^{\infty}(x^{2*m+1} * ((2*m+1)!)^{-1} * (-1)^m)$
$cos(x) = \sum_{m=0}^{\infty}(x^{2*m} * ((2*m)!)^{-1} * (-1)^m)$

zeta ise $\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow$
$\zeta(x_1 + i * x_2) =$
$(x_1^2 - 2*x_1 + 1 + x_2^2)^{-1} * \sum_{n=1}^{\infty}($
$\sum_{m=0}^{\infty}(ln(n*(2*m+1)) * x_2^{2*m+1} * ((2*m+1)!)^{-1} * (-1)^m) * x_2 * (n-1) * n^{-x_1} +$
$\sum_{m=0}^{\infty}(ln(n*2*m) * x_2^{2*m} * ((2*m)!)^{-1} * (-1)^m) * (x_1^2 - x_1 - n*x_1 + n + x_2^2) * n^{-x_1} -$
$\sum_{m=0}^{\infty}(ln((n+1)*(2*m+1)) * x_2^{2*m+1} * ((2*m+1)!)^{-1} * (-1)^m) * x_2 * n * (n+1)^{-x_1} -$
$\sum_{m=0}^{\infty}(ln((n+1)*2*m) * x_2^{2*m} * ((2*m)!)^{-1} * (-1)^m) * (1-x_1) * n * (n+1)^{-x_1}$
)
$+ i * (x_1^2 - 2*x_1 + 1 + x_2^2)^{-1} * \sum_{n=1}^{\infty}($
$(sin(x_2 * ln(n) + 1*(2\pi)/4) * x_2 * (n-1) + sin(x_2 * ln(n) + 2*(2\pi)/4) * (x_1^2 - x_1 - n*x_1 + n + x_2^2)) * n^{-x_1} -$
$(sin(x_2 * ln(n+1) + 1*(2\pi)/4) * x_2 + sin(x_2 * ln(n+1) + 2*(2\pi)/4) * (1-x_1)) * n * (n+1)^{-x_1}$
)
=
$(x_1^2 - 2*x_1 + 1 + x_2^2)^{-1} * \sum_{n=1}^{\infty}(\sum_{m=0}^{\infty}(x_2^{2*m} * (-1)^m * ((2*m)!)^{-1} * ($
$ln(n*(2*m+1)) * (2*m+1)^{-1} * x_2^2 * (n-1) * n^{-x_1} +$
$ln(n*2*m) * (x_1^2 - x_1 - n*x_1 + n + x_2^2) * n^{-x_1} -$
$ln((n+1)*(2*m+1)) * (2*m+1)^{-1} * x_2^2 * n * (n+1)^{-x_1} -$
$ln((n+1)*2*m) * (1-x_1) * n * (n+1)^{-x_1}$
)))
$+ i * (x_1^2 - 2*x_1 + 1 + x_2^2)^{-1} * \sum_{n=1}^{\infty}($
$(sin(x_2 * ln(n) + 1*(2\pi)/4) * x_2 * (n-1) + sin(x_2 * ln(n) + 2*(2\pi)/4) * (x_1^2 - x_1 - n*x_1 + n + x_2^2)) * n^{-x_1} -$
$(sin(x_2 * ln(n+1) + 1*(2\pi)/4) * x_2 + sin(x_2 * ln(n+1) + 2*(2\pi)/4) * (1-x_1)) * n * (n+1)^{-x_1}$

)
=))

2 zetafunktsiooni nullkohad

Võrdsustan zetafunktsiooni kuju zeta_funktsioon_2_q_ga, mille mujal tuletasin 0iga.

$$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_1 > 0 \wedge x_2 \in R \wedge x_2 \neq 0 \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow$$

$$\lim_{M\to\infty}($$
$$\sum_{n=1}^{floor(e^{(floor(M)*sgn(x_2)+q_1)*2\pi/x_2})}(sin(x_2 * ln(n) + (2\pi) * 1/4) * n^{-x_1})$$
$$-(sin((q_1 + 1/4) * 2\pi) * (1 - x_1) + sin((q_1 + 0/4) * 2\pi) * x_2) * e^{(floor(M)*sgn(x_2)+q_1)*2\pi*(1-x_1)/x_2} * ((1 - x_1)^2 + x_2^2)^{-1}$$
$$)$$
$$+ i * \lim_{M\to\infty}($$
$$\sum_{n=1}^{floor(e^{(floor(M)*sgn(x_2)+q_2)*2\pi/x_2})}(sin(x_2 * ln(n) + (2\pi) * 2/4) * n^{-x_1})$$
$$-(sin((q_2 + 2/4) * 2\pi) * (1 - x_1) + sin((q_2 + 1/4) * 2\pi) * x_2) * e^{(floor(M)*sgn(x_2)+q_2)*2\pi*(1-x_1)/x_2} * ((1 - x_1)^2 + x_2^2)^{-1}$$
$$) = 0$$
$$))))$$

kuna nii reaalosa kui imaginaarosa peavad nullid olema, siis võin reaalosa suvalise konstandiga läbi korrutada.

$$\forall_{q_1}(\forall_{q_2}(\forall_{Q_1}(\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_1 > 0 \wedge x_2 \in R \wedge x_2 \neq 0 \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow$$

$$Q_1 * \lim_{M\to\infty}($$
$$\sum_{n=1}^{floor(e^{(floor(M)*sgn(x_2)+q_1)*2\pi/x_2})}(sin(x_2 * ln(n) + (2\pi) * 1/4) * n^{-x_1})$$
$$-(sin((q_1 + 1/4) * 2\pi) * (1 - x_1) + sin((q_1 + 0/4) * 2\pi) * x_2) * e^{(floor(M)*sgn(x_2)+q_1)*2\pi*(1-x_1)/x_2} * ((1 - x_1)^2 + x_2^2)^{-1}$$
$$) +$$
$$\lim_{M\to\infty}($$
$$\sum_{n=1}^{floor(e^{(floor(M)*sgn(x_2)+q_2)*2\pi/x_2})}(sin(x_2 * ln(n) + (2\pi) * 2/4) * n^{-x_1})$$
$$-(sin((q_2 + 2/4) * 2\pi) * (1 - x_1) + sin((q_2 + 1/4) * 2\pi) * x_2) * e^{(floor(M)*sgn(x_2)+q_2)*2\pi*(1-x_1)/x_2} * ((1 - x_1)^2 + x_2^2)^{-1}$$
$$) = 0$$
$$)))))$$

viin konsatndi sulguse sisse.

$$\forall_{q_1}(\forall_{q_2}(\forall_{Q_1}(\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_1 > 0 \wedge x_2 \in R \wedge x_2 \neq 0 \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow$$

$$\lim_{M\to\infty}($$
$$Q_1 * \sum_{n=1}^{floor(e^{(floor(M)*sgn(x_2)+q_1)*2\pi/x_2})}(sin(x_2 * ln(n) + (2\pi) * 1/4) * n^{-x_1})$$
$$-Q_1 * (sin((q_1 + 1/4) * 2\pi) * (1 - x_1) + sin((q_1 + 0/4) * 2\pi) * x_2) * e^{(floor(M)*sgn(x_2)+q_1)*2\pi*(1-x_1)/x_2} * ((1 - x_1)^2 + x_2^2)^{-1} +$$
$$\sum_{n=1}^{floor(e^{(floor(M)*sgn(x_2)+q_2)*2\pi/x_2})}(sin(x_2 * ln(n) + (2\pi) * 2/4) * n^{-x_1})$$
$$-(sin((q_2 + 2/4) * 2\pi) * (1 - x_1) + sin((q_2 + 1/4) * 2\pi) * x_2) * e^{(floor(M)*sgn(x_2)+q_2)*2\pi*(1-x_1)/x_2} * ((1 - x_1)^2 + x_2^2)^{-1}$$
$$) = 0$$
$$)))))$$

muudan kirjutan ümber viies ühiseid kordajaid sulguse ette. eeldan siin, et summa ruuremast arvust väikseamani on sama, mi negatiivne summa väiksemast arvust suuremani $\sum_{n=a}^{b}(f(n)) = -\sum_{n=b+1}^{a-1}(f(n))$

$$\forall_{q_1}(\forall_{q_2}(\forall_{Q_1}(\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_1 > 0 \wedge x_2 \in R \wedge x_2 \neq 0 \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow$$
$$\lim_{M\to\infty}($$
$$\sum_{n=1}^{floor(e^{(floor(M)*sgn(x_2)+q_1)*2\pi/x_2})}(Q_1 * sin(x_2 * ln(n) + (2\pi) * 1/4) * n^{-x_1} + sin(x_2 * ln(n) + (2\pi) * 2/4) * n^{-x_1})$$
$$+\sum_{n=floor(e^{(floor(M)*sgn(x_2)+q_1)*2\pi/x_2})+1}^{floor(e^{(floor(M)*sgn(x_2)+q_2)*2\pi/x_2})}(sin(x_2 * ln(n) + (2\pi) * 2/4) * n^{-x_1})$$
$$-Q_1 * (sin((q_1 + 1/4) * 2\pi) * (1 - x_1) + sin((q_1 + 0/4) * 2\pi) * x_2) * e^{(floor(M)*sgn(x_2)+q_1)*2\pi*(1-x_1)/x_2} * ((1 - x_1)^2 + x_2^2)^{-1} - (sin((q_2 + 2/4) * 2\pi) * (1 - x_1) + sin((q_2 + 1/4) * 2\pi) * x_2) * e^{(floor(M)*sgn(x_2)+q_2)*2\pi*(1-x_1)/x_2} * ((1 - x_1)^2 + x_2^2)^{-1}) = 0$$

$$))))))$$

siinuse ja koosinuse summa saab ühiseks siinuseks teha nii, et argumendile on midagi liidetud.

$$\forall_{q_1}(\forall_{q_2}(\forall_{Q_1}(\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_1 > 0 \wedge x_2 \in R \wedge x_2 \neq 0 \wedge Q_1 \in R \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow$$
$$\lim_{M \to \infty}($$
$$\sum_{n=1}^{floor(e^{(floor(M)*sgn(x_2)+q_1)*2\pi/x_2})}(\sqrt{Q_1^2+1} * -sin(x_2 * ln(n) + arctan(-Q_1)) * n^{-x_1})$$
$$+ \sum_{n=floor(e^{(floor(M)*sgn(x_2)+q_1)*2\pi/x_2})+1}^{floor(e^{(floor(M)*sgn(x_2)+q_2)*2\pi/x_2})}(sin(x_2 * ln(n) + (2\pi) * 2/4) * n^{-x_1})$$
$$- Q_1 * (sin((q_1 + 1/4) * 2\pi) * (1 - x_1) + sin((q_1 + 0/4) * 2\pi) * x_2) * e^{(floor(M)*sgn(x_2)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 +$$
$$x_2^2)^{-1} - (sin((q_2 + 2/4) * 2\pi) * (1 - x_1) + sin((q_2 + 1/4) * 2\pi) * x_2) * e^{(floor(M)*sgn(x_2)+q_2)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 +$$
$$x_2^2)^{-1}) = 0$$
$$))))))$$

arctan(-x)=-arctan(x)

$$\forall_{q_1}(\forall_{q_2}(\forall_{Q_1}(\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_1 > 0 \wedge x_2 \in R \wedge x_2 \neq 0 \wedge Q_1 \in R \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow$$
$$\lim_{M \to \infty}($$
$$\sum_{n=1}^{floor(e^{(floor(M)*sgn(x_2)+q_1)*2\pi/x_2})}(\sqrt{Q_1^2+1} * -sin(x_2 * ln(n) - arctan(Q_1)) * n^{-x_1})$$
$$+ \sum_{n=floor(e^{(floor(M)*sgn(x_2)+q_1)*2\pi/x_2})+1}^{floor(e^{(floor(M)*sgn(x_2)+q_2)*2\pi/x_2})}(sin(x_2 * ln(n) + (2\pi) * 2/4) * n^{-x_1})$$
$$- Q_1 * (sin((q_1 + 1/4) * 2\pi) * (1 - x_1) + sin((q_1 + 0/4) * 2\pi) * x_2) * e^{(floor(M)*sgn(x_2)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1}$$
$$- (sin((q_2 + 2/4) * 2\pi) * (1 - x_1) + sin((q_2 + 1/4) * 2\pi) * x_2) * e^{(floor(M)*sgn(x_2)+q_2)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1}$$
$$) = 0$$
$$))))))$$

lihtsustan.

$$\forall_{q_1}(\forall_{q_2}(\forall_{Q_1}(\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_1 > 0 \wedge x_2 \in R \wedge x_2 \neq 0 \wedge Q_1 \in R \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow$$
$$\lim_{M \to \infty}($$
$$\sum_{n=1}^{floor(e^{(floor(M)*sgn(x_2)+q_1)*2\pi/x_2})}(\sqrt{Q_1^2+1} * -sin(x_2 * ln(n) - arctan(Q_1)) * n^{-x_1})$$
$$+ \sum_{n=floor(e^{(floor(M)*sgn(x_2)+q_1)*2\pi/x_2})+1}^{floor(e^{(floor(M)*sgn(x_2)+q_2)*2\pi/x_2})}(sin(x_2 * ln(n) + (2\pi) * 2/4) * n^{-x_1})$$
$$- e^{floor(M)*sgn(x_2)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1/2} * (Q_1 * e^{q_1*2\pi*(1-x_1)/x_2} * sin(q_1 * 2\pi + atan2(1-x_1, x_2))$$
$$+ e^{q_2*2\pi*(1-x_1)/x_2} * cos(q_2 * 2\pi + atan2(1-x_1, x_2)))$$
$$) = 0$$
$$))))))$$

korrutan -1ega läbi

$$\forall_{q_1}(\forall_{q_2}(\forall_{Q_1}(\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_1 > 0 \wedge x_2 \in R \wedge x_2 \neq 0 \wedge Q_1 \in R \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow$$
$$\lim_{M \to \infty}($$
$$\sum_{n=1}^{floor(e^{(floor(M)*sgn(x_2)+q_1)*2\pi/x_2})}(\sqrt{Q_1^2+1} * sin(x_2 * ln(n) - arctan(Q_1)) * n^{-x_1})$$
$$- \sum_{n=floor(e^{(floor(M)*sgn(x_2)+q_1)*2\pi/x_2})+1}^{floor(e^{(floor(M)*sgn(x_2)+q_2)*2\pi/x_2})}(sin(x_2 * ln(n) + (2\pi) * 2/4) * n^{-x_1})$$
$$+ e^{floor(M)*sgn(x_2)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1/2} * ($$
$$Q_1 * e^{q_1*2\pi*(1-x_1)/x_2} * sin(q_1 * 2\pi + atan2(1-x_1, x_2))$$
$$+ e^{q_2*2\pi*(1-x_1)/x_2} * cos(q_2 * 2\pi + atan2(1-x_1, x_2))$$
$$)$$
$$) = 0$$
$$))))))$$

ühe summadest saab asendada ilma summata avalisega. Kuna iga summa element annab lõpmatult väikese panuse, siis summa nagu integral.

$$\forall_{q_1}(\forall_{q_2}(\forall_{Q_1}(\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_1 > 0 \wedge x_2 \in R \wedge x_2 \neq 0 \wedge Q_1 \in R \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow$$
$$\lim_{M \to \infty}($$
$$\sum_{n=1}^{floor(e^{(floor(M)*sgn(x_2)+q_1)*2\pi/x_2})}(\sqrt{Q_1^2+1} * sin(x_2 * ln(n) - arctan(Q_1)) * n^{-x_1})$$
$$- (e^{(floor(M)*sgn(x_2)+q_2)*2\pi*(1-x_1)/x_2} * cos(q_2 * 2\pi + atan2(1-x_1, x_2))$$
$$- e^{(floor(M)*sgn(x_2)+q_1)*2\pi*(1-x_1)/x_2} * cos(q_1 * 2\pi + atan2(1-x_1, x_2))) * ((1-x_1)^2 + x_2^2)^{-1/2}$$
$$+ e^{floor(M)*sgn(x_2)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1/2} * ($$

$Q_1 * e^{q_1*2\pi*(1-x_1)/x_2} * sin(q_1*2\pi + atan2(1-x_1,x_2))$
$+ e^{q_2*2\pi*(1-x_1)/x_2} * cos(q_2*2\pi + atan2(1-x_1,x_2))$
$)$
$) = 0$
$)))))$

lihtsustan. q2 taandub välja.

$\forall_{q_1}(\forall_{Q_1}(\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_1 > 0 \wedge x_2 \in R \wedge x_2 \neq 0 \wedge Q_1 \in R \wedge \zeta(x_1 + i*x_2) = 0 \rightarrow$
$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)*sgn(x_2)+q_1)*2\pi/x_2})} (sin(x_2 * ln(n) - arctan(Q_1)) * n^{-x_1})$
$+ e^{(floor(M)*sgn(x_2)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1/2} * sin(q_1*2\pi + atan2(1-x_1,x_2) + 2\pi/4 - arctan(Q_1))$
$) = 0$
$))))$

arctan(Q_1) võtab väärtuseid -2pi/4 kuni 2pi/4. Ja valemis on Q_1 ainult arctan argumen, siis või arctan(Q_1) asemele panna arvu, mis on vahemikus -2pi/4 kuni 2pi/4.

$\forall_{q_1}(\forall_{Q_1}(\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_1 > 0 \wedge x_2 \in R \wedge x_2 \neq 0 \wedge Q_1 \in R \wedge Q_1 > -2\pi/4 \wedge Q_1 < 2\pi/4 \wedge \zeta(x_1 + i*x_2) = 0 \rightarrow$
$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)*sgn(x_2)+q_1)*2\pi/x_2})} (sin(x_2 * ln(n) - Q_1) * n^{-x_1})$
$+ e^{(floor(M)*sgn(x_2)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1/2} * sin(q_1*2\pi + atan2(1-x_1,x_2) + 2\pi/4 - Q_1)$
$) = 0$
$))))$

Panen Q_1 asemele samanimeline muutuja -Q_1.

$\forall_{q_1}(\forall_{Q_1}(\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_1 > 0 \wedge x_2 \in R \wedge x_2 \neq 0 \wedge Q_1 \in R \wedge Q_1 > -2\pi/4 \wedge Q_1 < 2\pi/4 \wedge \zeta(x_1 + i*x_2) = 0 \rightarrow$
$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)*sgn(x_2)+q_1)*2\pi/x_2})} (sin(x_2 * ln(n) + Q_1) * n^{-x_1})$
$+ e^{(floor(M)*sgn(x_2)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1/2} * sin(q_1*2\pi + atan2(1-x_1,x_2) + 2\pi/4 + Q_1)$
$) = 0$
$))))$

Q_1 võib olla ükskõik milline reaalarv, sest mistahes Q_1 puhul saan arvutada arvud k ja r $k = floor((Q_1 + 2\pi/4)/(2\pi/2))$; $r = Q_1 - k*2\pi/2$nii, et $sin(A+Q_1) = (-1)^k * sin(A+r)$. kuna mälemad siinused muudavad k muutudes korraga märki, siis võrrandi täesus see märgi muutmine ei mõjuta ja r on alati algses vahemikus -2pi/4 kuni 2pi/4. kui $Q_1 = -2\pi/4$, siis $sin(A - 2\pi/4) = sin(A) - \sqrt{2} * sin(A + 2\pi/8)$:

$\forall_{q_1}(\forall_{Q_1}(\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_1 > 0 \wedge x_2 \in R \wedge x_2 \neq 0 \wedge Q_1 \in R \wedge \zeta(x_1 + i*x_2) = 0 \rightarrow$
$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)*sgn(x_2)+q_1)*2\pi/x_2})} (sin(x_2 * ln(n) + Q_1) * n^{-x_1})$
$+ e^{(floor(M)*sgn(x_2)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1/2} * sin(q_1*2\pi + atan2(1-x_1,x_2) + 2\pi/4 + Q_1)$
$) = 0$
$))))$

muutuja Q_1 asemele $Q_1/(2\pi)$ valem zeta_0koht_1:

$\forall_{q_1}(\forall_{Q_1}(\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_1 > 0 \wedge x_2 \in R \wedge x_2 \neq 0 \wedge Q_1 \in R \wedge \zeta(x_1 + i*x_2) = 0 \rightarrow$
$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)*sgn(x_2)+q_1)*2\pi/x_2})} (sin(x_2 * ln(n) + Q_1 * (2\pi)) * n^{-x_1})$
$+ e^{(floor(M)*sgn(x_2)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1/2} * sin(q_1*2\pi + atan2(1-x_1,x_2) + 2\pi/4 + Q_1 * (2\pi))$
$) = 0$
$))))$

võtan q_1 asemel kasutusele uue muutuja p_1. $q_1 = p_1 * x_2/(2\pi) - floor(M) * sgn(x_2)$. nimetan Q_1 ümber nimega p_2.

$\forall_{q_1}(\forall_{Q_1}(\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_1 > 0 \wedge x_2 \in R \wedge x_2 \neq 0 \wedge Q_1 \in R \wedge \zeta(x_1 + i*x_2) = 0 \rightarrow$
$\lim_{p_1\to\infty}($
$\sum_{n=1}^{floor(e^{p_1})} (sin(x_2 * ln(n) + p_2 * (2\pi)) * n^{-x_1})$
$+ e^{p_1*(1-x_1)} * ((1-x_1)^2 + x_2^2)^{-1/2} * sin(p_1 * x_2 + atan2(1-x_1,x_2) + 2\pi/4 + Q_1 * (2\pi))$
$) = 0$

))))

2.1 tagumine pool nulliks

kuna valem valem zeta_0koht_1 kehtib iga q_1 väärtuse korral saan q_1 väärtuseks valida $q_1 = -Q_1 - 1/4 - atan2(1 - x_1, x_2)/(2\pi)$. Sel juhul $sin(q_1 * 2\pi + atan2(1 - x_1, x_2) + 2\pi/4 + Q_1 * (2\pi)) = 0$ seega:

$\forall_{Q_1}(\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_1 > 0 \wedge x_2 \in R \wedge x_2 \neq 0 \wedge Q_1 \in R \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow$
$\lim_{M\to\infty}(\sum_{n=1}^{floor(e^{((floor(M)*sgn(x_2)-Q_1-1/4)*2\pi-atan2(1-x_1,x_2))/x_2})}(sin(x_2 * ln(n) + Q_1 * (2\pi)) * n^{-x_1})) = 0$
)))

2.2 q=+-1/4

kuna valem valem zeta_0koht_1 kehtib iga q_1 väärtuse korral saan q_1 väärtuseks valida q_1=-1/4-Q_1 või q_1=1/4-Q_1. Sel juhul sin((q_1+Q_1+1/4)*2\pi)=0 seega:

$\forall_{Q_1}(\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_1 > 0 \wedge x_2 \in R \wedge x_2 \neq 0 \wedge Q_1 \in R \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow$
$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)*sgn(x_2)-1/4-Q_1)*2\pi/x_2})}(sin(x_2 * ln(n) + Q_1 * (2\pi)) * n^{-x_1})$
$+ e^{(floor(M)*sgn(x_2)-1/4-Q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1/2} * sin((-1/4-Q_1) * 2\pi + atan2(1-x_1, x_2) + 2\pi/4 + Q_1 * (2\pi))$
$) = 0$
$\wedge$
$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)*sgn(x_2)+1/4-Q_1)*2\pi/x_2})}(sin(x_2 * ln(n) + Q_1 * (2\pi)) * n^{-x_1})$
$+ e^{(floor(M)*sgn(x_2)+1/4-Q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1/2} * sin((1/4-Q_1) * 2\pi + atan2(1-x_1, x_2) + 2\pi/4 + Q_1 * (2\pi))$
$) = 0$
)))

samahästi võib m'i 2 korda aeglasemalt suurendada, mis tõttu need 2 varianti muutuvad samasuguseks.

$\forall_{Q_1}(\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_1 > 0 \wedge x_2 \in R \wedge x_2 \neq 0 \wedge Q_1 \in R \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow$
$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)*sgn(x_2)/2+1/4-Q_1)*2\pi/x_2})}(sin(x_2 * ln(n) + Q_1 * (2\pi)) * n^{-x_1})$
$+ e^{(floor(M)*sgn(x_2)/2+1/4-Q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1/2} * sin((floor(M) * sgn(x_2)/2 + 1/4 - Q_1) * 2\pi + atan2(1-x_1, x_2) + 2\pi/4 + Q_1 * (2\pi))$
$) = 0$
)))

ja siis jälle Mi 2 korda kiiremini suurendada.

$\forall_{Q_1}(\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_1 > 0 \wedge x_2 \in R \wedge x_2 \neq 0 \wedge Q_1 \in R \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow$
$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)*sgn(x_2)+1/4-Q_1)*2\pi/x_2})}(sin(x_2 * ln(n) + Q_1 * (2\pi)) * n^{-x_1})$
$+ e^{(floor(M)*sgn(x_2)+1/4-Q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1/2} * sin((1/4-Q_1) * 2\pi + atan2(1-x_1, x_2) + 2\pi/4 + Q_1 * (2\pi))$
$) = 0$
)))

2.3 q=+-1/8

ei tea kas vaja

3 muu

3.0.1 c

$lim_{M\to\infty}(M*(M+1)^{-x_1}-M^{1-x_1})=$
$x_1>=0\to return(0)$
$x_1<0\to return(\infty)$

3.0.2 b

$lim_{M\to\infty}($
$(sin(x_2*ln(M+1)+2*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M+1)+1*(2\pi)/4)*x_2)*M*(M+1)^{-x_1}*(x_1^2-2*x_1+$
$x_2^2+1)^{-1}-$
$(sin(x_2*ln(M+1)+2*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M+1)+1*(2\pi)/4)*x_2)*M^{1-x_1}*(x_1^2-2*x_1+x_2^2+1)^{-1}$
$)=$
$lim_{M\to\infty}((sin(x_2*ln(M+1)+2*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M+1)+1*(2\pi)/4)*x_2)*$
$(M*(M+1)^{-x_1}-M^{1-x_1})$
$)=$
$x_1>0\to return(0)$
$x_1<=0\to return(\infty)$

3.0.3 a

$lim_{M\to\infty}($
$(sin(x_2*ln(M+1)+2*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M+1)+1*(2\pi)/4)*x_2)*M*(M+1)^{-x_1}*(x_1^2-2*x_1+$
$x_2^2+1)^{-1}-$
$(sin(x_2*ln(M)+2*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M)+1*(2\pi)/4)*x_2)*M^{1-x_1}*(x_1^2-2*x_1+x_2^2+1)^{-1}$
$)=$
$lim_{M\to\infty}($
$(sin(x_2*ln(M+1)+2*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M+1)+1*(2\pi)/4)*x_2)*M*(M+1)^{-x_1}-$
$(sin(x_2*ln(M)+2*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M)+1*(2\pi)/4)*x_2)*M^{1-x_1}$
$)=$

3.0.4 d

$lim_{M\to\infty}($
$(sin(x_2*ln(M+1)+2*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M+1)+1*(2\pi)/4)*x_2)*M^{1-x_1}*(x_1^2-2*x_1+x_2^2+1)^{-1}-$
$(sin(x_2*ln(M)+2*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M)+1*(2\pi)/4)*x_2)*M^{1-x_1}*(x_1^2-2*x_1+x_2^2+1)^{-1}$
$)=$

$(x_1^2-2*x_1+x_2^2+1)^{-1}*lim_{M\to\infty}(M^{1-x_1}*($
$(1-x_1)*(sin(x_2*ln(M+1)+2*(2\pi)/4)-sin(x_2*ln(M)+2*(2\pi)/4))+$
$x_2*(sin(x_2*ln(M+1)+1*(2\pi)/4)-sin(x_2*ln(M)+1*(2\pi)/4))$
$))=$

$(x_1^2-2*x_1+x_2^2+1)^{-1}*lim_{M\to\infty}(M^{1-x_1}*($
$(cos(x_2*ln(M+1))-cos(x_2*ln(M)))*x_2-$
$(sin(x_2*ln(M+1))-sin(x_2*ln(M)))*(1-x_1)))$

$lim_{M\to\infty}(M^{1-x_1}*($
$abs((cos(x_2*ln(M+1))-cos(x_2*ln(M)))*x_2))+$
$abs((sin(x_2*ln(M+1))-sin(x_2*ln(M)))*(1-x_1)))$
$))$

$lim_{M\to\infty}(M^{1-x_1} * (
abs((x_2 * ln(M+1) - x_2 * ln(M)) * x_2) +
abs((x_2 * ln(M+1) - x_2 * ln(M)) * (1-x_1))
))$

$lim_{M\to\infty}(M^{1-x_1} * (
abs((ln(M+1) - ln(M)) * x_2^2) +
abs((ln(M+1) - ln(M)) * (1-x_1) * x_2)
))$

$lim_{M\to\infty}(M^{1-x_1} * (
(ln(M+1) - ln(M)) * x_2^2 +
(ln(M+1) - ln(M)) * abs((1-x_1) * x_2)
))$

$lim_{M\to\infty}(M^{1-x_1} * (
(ln(M+1) - ln(M)) * (abs(x_2) * (abs(x_2) + abs(1-x_1)))
))$

$lim_{M\to\infty}(M^{1-x_1} * (ln(M+1) - ln(M)) * (abs(x_2) * (abs(x_2) + abs(1-x_1))))$

$lim_{M\to\infty}(M^{1-x_1} * (ln(M+1) - ln(M)))$

$lim_{M\to\infty}(M^{-x_1} * \ (ln(M+1) - ln(M)) * M)$

$lim_{M\to\infty}(M^{-x_1})$

$x_1 > 0 \to return\ 0$

3.0.5 f

$ln(M) * x_2 = (k * 2\pi)/2$

$M = e^{(k*2\pi)/(2*x_2)} = e^{\frac{k*2\pi}{2*x_2}}$

$k = 2 * ln(M) * x_2/(2\pi) = \frac{2*ln(M)*x_2}{2\pi}$

$x_2 = k * 2\pi/ln(M)/2 = \frac{k*2\pi}{ln(M)*2}$

asendus

$M_{uus} = e^{(floor(M)+q)*2\pi/x_2} = e^{\frac{floor(M)*2\pi}{x_2}} = e^{(floor(M)*2\pi+\alpha)/x_2}$

asendus2

$ln(M) * x_2 = (k * 2\pi) + \alpha$

M=$e^{(k*2\pi+\alpha)/(x_2)} = e^{\frac{k*2\pi+\alpha}{x_2}}$

k=(ln(M)*$x_2 - \alpha)/(2\pi) = \frac{ln(M)*x_2-\alpha}{2\pi}$

$x_2 = ((k * 2\pi) + \alpha)/ln(M)$

q=$\alpha/2\pi$

3.0.6 g

$sin(ln(x)) =$

$\forall_x(x > -1 \wedge x < 1 \rightarrow$
$ln(x+1) = \sum_{n=1}^{\infty}((-1)^{n+1} * n^{-1} * x^n)$
)

muutuja vahetus x-1

$\forall_x(x > 0 \wedge x < 2 \rightarrow$
$ln(x) = \sum_{n=1}^{\infty}((-1)^{n+1} * n^{-1} * (x-1)^n)$
)

mv 1/x

$\forall_x(x > 1/2 \rightarrow$
$ln(1/x) = \sum_{n=1}^{\infty}((-1)^{n+1} * n^{-1} * (1/x-1)^n)$
)

mv summas

$\forall_x(x > 1/2 \rightarrow$
$ln(1/x) = \sum_{n=1}^{\infty}((-1)^{n+1} * n^{-1} * (1/x-1)^n)$
)

..

3.0.7 ln maclaurin

https://math.stackexchange.com/questions/585154/taylor-series-for-logx

valem ln_maclaurin:

$\forall_x(x > 0 \rightarrow$
$ln(x) = 2 * \sum_{k=0}^{\infty}((2*k+1)^{-1} * (x-1)^{2*k+1} * (x+1)^{-2*k-1})$
)

3.0.8 sin(ln(x)) maclaurin

sin:

$\forall_x(x \in R \rightarrow$
$sin(x) = \sum_{k=0}^{\infty}((-1)^k * x^{2*k+1} * \Gamma(2*k+2)^{-1})$
)

sin(y*ln(x)): Valem sin_y_ln_x_maclaurin_3:

$\forall_x(x > 0 \rightarrow$
$sin(y*ln(x)) = \sum_{k_1=0}^{\infty}((\sum_{k_2=0}^{\infty}((2*k_2+1)^{-1} * (x-1)^{2*k_2+1} * (x+1)^{-2*k_2-1}))^{2*k_1+1} * y^{2*k_1+1} * factorial(2*k_1+1)^{-1} *$
$2^{2*k_1+1} * (-1)^{k_1})$
)

kasutan Multinomial theoreemi. Siin on eeldatud, et True'ga korrutamine on nagu 1ega korrutamine ja Falsega korrutamine on nagu 0'iga korrutamine. Valem sin_y_ln_x_multinomial_1:

$\forall_x(x > 0 \rightarrow$
$sin(y*ln(x)) = lim_{M\rightarrow\infty}(\sum_{k_1=0}^{M-1}(\sum_{k_2=0}^{(2*k_1+2)^M-1}($
$(\sum_{k_3=0}^{M-1}(floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2)\ ) = 2*k_1+1) *$
$\Pi_{k_3=0}^{M-1}(factorial(\ floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2)\ )^{-1} * ((2*k_3+1)^{-1} * (x-1)^{2*k_3+1} * (x+1)^{-2*k_3-1})^{floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2)})$
$) * y^{2*k_1+1} * 2^{2*k_1+1} * (-1)^{k_1}))$
)

elementide korrutamise asemel võib nende astemed liita:

$\forall_x(x > 0 \rightarrow sin(y*ln(x)) =$
$lim_{M\rightarrow\infty}(\sum_{k_1=0}^{M-1}(\sum_{k_2=0}^{(2*k_1+2)^M-1}($
$(\sum_{k_3=0}^{M-1}(floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2)\ ) = 2*k_1+1) *$
$\Pi_{k_3=0}^{M-1}(factorial(floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2))^{-1} * (2*k_3+1)^{-(floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2))}) * ((x-1)/(x+$
$1))^{\sum_{k_3=0}^{M-1}((2*k_3+1)*(floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2)))} * y^{2*k_1+1} * 2^{2*k_1+1} * (-1)^{k_1})))$
)

lihtsustan. Valem sin_y_ln_x_multinomial:

$$\forall_x(x>0 \to sin(y*ln(x)) =$$
$$lim_{M\to\infty}(\sum_{k_1=0}^{M-1}(y^{2*k_1+1}*2^{2*k_1+1}*(-1)^{k_1}*\sum_{k_2=0}^{(2*k_1+2)^M-1}($$
$$(\sum_{k_3=0}^{M-1}(\ floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2)\ ) = 2*k_1+1)*$$
$$(x-1)^{\sum_{k_3=0}^{M-1}((2*k_3+1)*(floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2)))}*$$
$$(x+1)^{-\sum_{k_3=0}^{M-1}((2*k_3+1)*(floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2)))}*$$
$$\Pi_{k_3=0}^{M-1}(factorial(floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2))^{-1}*(2*k_3+1)^{-(floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2))}))))$$
$$)$$

viin teguri sulgude sisse:

$$\forall_x(x>0 \to sin(y*ln(x)) =$$
$$lim_{M\to\infty}(\sum_{k_1=0}^{M-1}(\sum_{k_2=0}^{(2*k_1+2)^M-1}($$
$$(\sum_{k_3=0}^{M-1}(\ floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2)\ ) = 2*k_1+1)*$$
$$y^{2*k_1+1}*2^{2*k_1+1}*(-1)^{k_1}*$$
$$(x-1)^{\sum_{k_3=0}^{M-1}((2*k_3+1)*(floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2)))}*$$
$$(x+1)^{-\sum_{k_3=0}^{M-1}((2*k_3+1)*(floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2)))}*$$
$$\Pi_{k_3=0}^{M-1}(factorial(\ floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2)\ )^{-1}*(2*k_3+1)^{-(floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2))})$$
$$)))$$
$$)$$

selle asemel et üle eri n2'e ristsummade summeeridaarvutan lihtsalt n2'e ristsumma. Valem sin_y_ln_x_multinomial_2:

(2*k_2+1)^{-(floor(k_1/(2*M)^{k_2})\%(2*M))}

$$\forall_x(x>0 \to sin(y*ln(x)) =$$
$$lim_{M\to\infty}(\sum_{k_1=0}^{(2*M)^M-1}($$
$$(\sum_{k_2=0}^{M-1}(floor(k_1/(2*M)^{k_2})\%(2*M))\%2)*$$
$$(\sum_{k_2=0}^{M-1}(floor(k_1/(2*M)^{k_2})\%(2*M)) < 2*M)*$$
$$(y*2)^{\sum_{k_2=0}^{M-1}(floor(k_1/(2*M)^{k_2})\%(2*M))}*$$
$$(-1)^{(\sum_{k_2=0}^{M-1}(floor(k_1/(2*M)^{k_2})\%(2*M))-1)/2}*$$
$$(x-1)^{\sum_{k_2=0}^{M-1}((2*k_2+1)*(floor(k_1/(2*M)^{k_2})\%(2*M)))}*$$
$$(x+1)^{-\sum_{k_2=0}^{M-1}((2*k_2+1)*(floor(k_1/(2*M)^{k_2})\%(2*M)))}*$$
$$\Pi_{k_2=0}^{M-1}(factorial(floor(k_1/(2*M)^{k_2})\%(2*M))^{-1}*(2*k_2+1)^{-(floor(k_1/(2*M)^{k_2})\%(2*M))})$$
$$))$$
$$)$$

Valem sin_y_ln_x_multinomial_4:

$$\forall_x(x>0 \to sin(y*ln(x)) =$$
$$lim_{M\to\infty}(\sum_{k_1=0}^{(2*M)^M-1}($$
$$(\sum_{k_2=0}^{M-1}(floor(k_1/(2*M)^{k_2})\%(2*M))\%2)*$$
$$(y*2)^{\sum_{k_2=0}^{M-1}(floor(k_1/(2*M)^{k_2})\%(2*M))}*$$
$$(-1)^{(\sum_{k_2=0}^{M-1}(floor(k_1/(2*M)^{k_2})\%(2*M))+3)/2}*$$
$$(x-1)^{\sum_{k_2=0}^{M-1}((2*k_2+1)*(floor(k_1/(2*M)^{k_2})\%(2*M)))}*$$
$$(x+1)^{-\sum_{k_2=0}^{M-1}((2*k_2+1)*(floor(k_1/(2*M)^{k_2})\%(2*M)))}*$$
$$\Pi_{k_2=0}^{M-1}(factorial(floor(k_1/(2*M)^{k_2})\%(2*M))^{-1}*(2*k_2+1)^{-(floor(k_1/(2*M)^{k_2})\%(2*M))})$$
$$))$$
$$)$$

kasutan multimomial astendajategenereerimiseks teist valemit $v(A,k_2,M)=\sum_{k_3=0}^{\infty}(floor(k_2/2^{k_3*M+A}\%2)*2^{k_3})$. Ehk biti indexi jääk määrab, et mitmenda astendaja bit see on.

$$\forall_x(x>0 \to sin(y*ln(x)) =$$
$$lim_{M\to\infty}(\sum_{k_1=0}^{(2*M)^M-1}($$

$(\sum_{k_2=0}^{M-1}(\sum_{k_3=0}^{\infty}(floor(k_1/2^{k_3*M+k_2}\%2)*2^{k_3}))\%2)*$
$(y*2)^{\sum_{k_2=0}^{M-1}(\sum_{k_3=0}^{\infty}(floor(k_1/2^{k_3*M+k_2}\%2)*2^{k_3}))}*$
$(-1)^{(\sum_{k_2=0}^{M-1}(\sum_{k_3=0}^{\infty}(floor(k_1/2^{k_3*M+k_2}\%2)*2^{k_3}))+3)/2}*$
$(x-1)^{\sum_{k_2=0}^{M-1}((2*k_2+1)*(\sum_{k_3=0}^{\infty}(floor(k_1/2^{k_3*M+k_2}\%2)*2^{k_3})))}*$
$(x+1)^{-\sum_{k_2=0}^{M-1}((2*k_2+1)*(\sum_{k_3=0}^{\infty}(floor(k_1/2^{k_3*M+k_2}\%2)*2^{k_3})))}*$
$\Pi_{k_2=0}^{M-1}(factorial(\sum_{k_3=0}^{\infty}(floor(k_1/2^{k_3*M+k_2}\%2)*2^{k_3}))^{-1}*(2*k_2+1)^{-\sum_{k_3=0}^{\infty}(floor(k_1/2^{k_3*M+k_2}\%2)*2^{k_3})})$
$))$
$)$

3.0.9 cos(ln(x)) maclaurin

cos:
$\forall_x(x \in R \rightarrow$
$cos(x) = \sum_{k=0}^{\infty}((-1)^k * x^{2*k} * factorial(2*k)^{-1})$
$)$
cos(y*ln(x)):
$\forall_x(x > 0 \rightarrow$
$cos(y*ln(x)) = \sum_{k_1=0}^{\infty}((-1)^{k_1} * (2*\sum_{k_2=0}^{\infty}((2*k_2+1)^{-1} * (x-1)^{2*k_2+1} * (x+1)^{-2*k_2-1}))^{2*k_1} * y^{2*k_1} * \Gamma(2*k_1+1)^{-1})$
$)$
Valem cos_y_ln_x_multinomial:
$\forall_x(x > 0 \rightarrow cos(y*ln(x)) =$
$lim_{M\rightarrow\infty}(\sum_{k_1=0}^{M-1}(y^{2*k_1} * 2^{2*k_1} * (-1)^{k_1} * \sum_{k_2=0}^{(2*k_1+1)^M-1}($
$(\sum_{k_3=0}^{M-1}(\ floor(k_2/(2*k_1+1)^{k_3})\%(2*k_1+1)\ ) = 2*k_1) *$
$(x-1)^{\sum_{k_3=0}^{M-1}((2*k_3+1)*(floor(k_2/(2*k_1+1)^{k_3})\%(2*k_1+1)))}*$
$(x+1)^{-\sum_{k_3=0}^{M-1}((2*k_3+1)*(floor(k_2/(2*k_1+1)^{k_3})\%(2*k_1+1)))}*$
$\Pi_{k_3=0}^{M-1}(factorial(\ floor(k_2/(2*k_1+1)^{k_3})\%(2*k_1+1)\ )^{-1} * (2*k_3+1)^{-floor(k_2/(2*k_1+1)^{k_3})\%(2*k_1+1)})$
$)))$
$)$
Valem cos_y_ln_x_multinomial_2:
$\forall_x(x > 0 \rightarrow cos(y*ln(x)) =$
$lim_{M\rightarrow\infty}(\sum_{k_1=0}^{(2*M-1)^M-1}($
$((1+\sum_{k_2=0}^{M-1}(floor(k_1/(2*M-1)^{k_2})\%(2*M-1)))\%2) *$
$(\sum_{k_2=0}^{M-1}(floor(k_1/(2*M-1)^{k_2})\%(2*M-1)) < 2*M-1) *$
$(y*2)^{\sum_{k_2=0}^{M-1}(floor(k_1/(2*M-1)^{k_2})\%(2*M-1))}*$
$(-1)^{\sum_{k_2=0}^{M-1}(floor(k_1/(2*M-1)^{k_2})\%(2*M-1))/2}*$
$(x-1)^{\sum_{k_2=0}^{M-1}((2*k_2+1)*(floor(k_1/(2*M-1)^{k_2})\%(2*M-1)))}*$
$(x+1)^{-\sum_{k_2=0}^{M-1}((2*k_2+1)*(floor(k_1/(2*M-1)^{k_2})\%(2*M-1)))}*$
$\Pi_{k_2=0}^{M-1}(factorial(floor(k_1/(2*M-1)^{k_2})\%(2*M-1))^{-1} * (2*k_2+1)^{-(floor(k_1/(2*M-1)^{k_2})\%(2*M-1))})$
$))$
$)$
positsioonsüsteemi, millega eksponendid on kodeeritud on nüüd 2*M-1 asemel 2*M. veel üks valem:
$\forall_x(x > 0 \rightarrow cos(y*ln(x)) =$
$lim_{M\rightarrow\infty}(\sum_{n_1=0}^{(2*M)^M-1}($
$((1+\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M)))\%2) *$
$(\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M)) < 2*M-1) *$
$(y*2)^{\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))}*$

$(-1)^{\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))/2} *$
$(x-1)^{\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_1/(2*M)^{n_2})\%(2*M)))} *$
$(x+1)^{-\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_1/(2*M)^{n_2})\%(2*M)))} *$
$\Pi_{n_2=0}^{M-1}(factorial(floor(n_1/(2*M)^{n_2})\%(2*M))^{-1} * (2*n_2+1)^{-(floor(n_1/(2*M)^{n_2})\%(2*M))})$
))
)

Valem cos_y_ln_x_multinomial_4_2:

$\forall_x(x>0 \rightarrow cos(y*ln(x)) =$
$lim_{M\rightarrow\infty}(\sum_{n_1=0}^{(2*M)^M-1}($
$((1+\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M)))\%2) * (y*2)^{\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))} *$
$(-1)^{\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))/2} *$
$(x-1)^{\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_1/(2*M)^{n_2})\%(2*M)))} *$
$(x+1)^{-\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_1/(2*M)^{n_2})\%(2*M)))} *$
$\Pi_{n_2=0}^{M-1}(factorial(floor(n_1/(2*M)^{n_2})\%(2*M))^{-1} * (2*n_2+1)^{-(floor(n_1/(2*M)^{n_2})\%(2*M))})$
))
)

3.0.10 sisemine summa

$A = \sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_0/(2*M)^{n_2})\%(2*M)))$

$-\sum_{n_1=0}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-1}($
$n_1^A *$
$(n_1+1)^{-x_1} *$
$(n_1+2)^{-A} *$
)

kasutan binomial valemit:$\forall_n(\forall_a(\forall_b((a+b)^n = \sum_{k=0}^{\infty}(faktorial(n)*faktorial(k)^{-1}*faktorial(n-k)^{-1}*a^k*b^{n-k})$

saadud tulemus:

$\sum_{n_1=0}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-1}($
$\sum_{k=0}^{\infty}(faktorial(-x_1)*faktorial(k)^{-1}*faktorial(-x_1-k)^{-1}*n_1^{k+A}) *$
$\sum_{k=0}^{\infty}(faktorial(-A)*faktorial(k)^{-1}*faktorial(-A-k)^{-1}*n_1^k*2^{-A-k}) *$
)

panen 1 summa 2 sisse:

$\sum_{n_1=0}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-1}($
$\sum_{k_1=0}^{\infty}(\sum_{k_2=0}^{\infty}(faktorial(-x_1)*faktorial(k_1)^{-1}*faktorial(-x_1-k_1)^{-1}*n_1^{k_1+A}*faktorial(-A)*faktorial(k_2)^{-1}*$
$faktorial(-A-k_2)^{-1}*n_1^{k_2}*2^{-A-k_2})) *$
)

lihtsustan seest ja viin n1 summa sisse:

$\sum_{k_1=0}^{\infty}(\sum_{k_2=0}^{\infty}(faktorial(-x_1)*faktorial(k_1)^{-1}*faktorial(-x_1-k_1)^{-1}*faktorial(-A)*faktorial(k_2)^{-1}*faktorial(-A-k_2)^{-1}*2^{-A-k_2}*$
$\sum_{n_1=0}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-1}(n_1^{k_1+k_2+A})$
))

3.0.11 veel

$\int_a^b (dx * sin(ln(x))) = (-1/2 * b * (cos(log(b)) - sin(log(b)))) - (-1/2 * a * (cos(log(a)) - sin(log(a)))) = (b/2 * (sin(ln(b)) - cos(ln(b)))) - (a/2 * (sin(ln(a)) - cos(ln(a)))) = 2^{-1/2} * (b * sin(ln(b) - 2\pi/8) - a * sin(ln(a) - 2\pi/8)) = 2^{-1/2} * (a * cos(ln(a)) - b * cos(ln(b)))$

veel

$\int_a^{a+b} (dx * sin(ln(x))) = 2^{-1/2} * (a * cos(ln(a)) - (a+b) * cos(ln(a+b))) = 2^{-1/2} * (a * (cos(ln(a)) - cos(ln(a+b))) - b * cos(ln(a+b)))$

veel

$\int_{floor(e^{(floor(M)/2+1/4)*2\pi/abs(x_2)-arctan((1-x_1)/x_2)/x_2})}^{floor(e^{(floor(M)/2+1/4)*2\pi/abs(x_2)-arctan((1-x_1)/x_2)/x_2})+b} (dx * sin(ln(x))) = 2^{-1/2} * (floor(e^{(floor(M)/2+1/4)*2\pi/abs(x_2)-arctan((1-x_1)/x_2)/x_2}) * (cos(ln(floor(e^{(floor(M)/2+1/4)*2\pi/abs(x_2)-arctan((1-x_1)/x_2)/x_2}))) - cos(ln(floor(e^{(floor(M)/2+1/4)*2\pi/abs(x_2)-arctan((1-x_1)/x_2)/x_2}) + b))) - b * cos(ln(floor(e^{(floor(M)/2+1/4)*2\pi/abs(x_2)-arctan((1-x_1)/x_2)/x_2}) + b))) = 2^{-1/2} * (floor(e^{(floor(M)/2+1/4)*2\pi/abs(x_2)-arctan((1-x_1)/x_2)/x_2}) * (cos((floor(M)/2 + 1/4) * 2\pi/abs(x_2) - arctan((1 - x_1)/x_2)/x_2) - cos(ln(floor(e^{(floor(M)/2+1/4)*2\pi/abs(x_2)-arctan((1-x_1)/x_2)/x_2}) + b))) - b * cos(ln(floor(e^{(floor(M)/2+1/4)*2\pi/abs(x_2)-arctan((1-x_1)/x_2)/x_2}) + b)))$

3.0.12 integral

siinuse oma $\int_A^B (dy * sin(x_2 * ln(y) + (2\pi) * 2/4) * y^{-x_1}) =$

$-\int_A^B (dy * sin(x_2 * ln(y)) * y^{-x_1}) =$

$\frac{B^{1-x_1} * ((x_1-1) * sin(x_2 * ln(B)) + x_2 * cos(x_2 * ln(B)))}{x_1^2 - 2 * x_1 + x_2^2 + 1} - \frac{A^{1-x_1} * ((x_1-1) * sin(x_2 * ln(A)) + x_2 * cos(x_2 * ln(A)))}{x_1^2 - 2 * x_1 + x_2^2 + 1} =$

$(B^{1-x_1} * ((x_1 - 1) * sin(x_2 * ln(B)) + x_2 * cos(x_2 * ln(B))) - A^{1-x_1} * ((x_1 - 1) * sin(x_2 * ln(A)) + x_2 * cos(x_2 * ln(A)))) * ((1 - x_1)^2 + x_2^2)^{-1} =$

$(B^{1-x_1} * (x_1 - 1) * sin(x_2 * ln(B)) + B^{1-x_1} * x_2 * cos(x_2 * ln(B)) - A^{1-x_1} * (x_1 - 1) * sin(x_2 * ln(A)) - A^{1-x_1} * x_2 * cos(x_2 * ln(A))) * ((1 - x_1)^2 + x_2^2)^{-1} =$

$(-B^{1-x_1} * (1 - x_1) * sin(x_2 * ln(B)) + B^{1-x_1} * x_2 * cos(x_2 * ln(B)) + A^{1-x_1} * (1 - x_1) * sin(x_2 * ln(A)) - A^{1-x_1} * x_2 * cos(x_2 * ln(A))) * ((1 - x_1)^2 + x_2^2)^{-1} =$

$(B^{1-x_1} * (-(1 - x_1) * sin(x_2 * ln(B)) + x_2 * cos(x_2 * ln(B))) + A^{1-x_1} * ((1 - x_1) * sin(x_2 * ln(A)) - x_2 * cos(x_2 * ln(A)))) * ((1 - x_1)^2 + x_2^2)^{-1} =$

$(B^{1-x_1} * \sqrt{((1 - x_1)^2 + x_2^2)} * sin(x_2 * ln(B) - arctan(x_2/(1 - x_1))) + A^{1-x_1} * \sqrt{((1 - x_1)^2 + x_2^2)} * sin(x_2 * ln(A) - arctan(x_2/(1 - x_1)))) * ((1 - x_1)^2 + x_2^2)^{-1} =$

$(B^{1-x_1} * sin(x_2 * ln(B) - arctan(x_2/(1 - x_1))) + A^{1-x_1} * sin(x_2 * ln(A) - arctan(x_2/(1 - x_1)))) * ((1 - x_1)^2 + x_2^2)^{-1/2}$

coosinuse oma $\int_A^B (dy * sin(x_2 * ln(y) + (2\pi) * 1/4) * y^{-x_1}) =$

$\int_A^B (dy * cos(x_2 * ln(y)) * y^{-x_1}) =$

$\frac{B^{1-x_1} * (x_2 * sin(x_2 * ln(B)) - (x_1 - 1) * cos(x_2 * ln(B)))}{x_1^2 - 2 * x_1 + x_2^2 + 1} - \frac{A^{1-x_1} * (x_2 * sin(x_2 * ln(A)) - (x_1 - 1) * cos(x_2 * ln(A)))}{x_1^2 - 2 * x_1 + x_2^2 + 1} =$

$(B^{1-x_1} * (x_2 * sin(x_2 * ln(B)) - (x_1 - 1) * cos(x_2 * ln(B))) - A^{1-x_1} * (x_2 * sin(x_2 * ln(A)) - (x_1 - 1) * cos(x_2 * ln(A)))) * ((1 - x_1)^2 + x_2^2)^{-1} =$

$(B^{1-x_1} * x_2 * sin(x_2 * ln(B)) - B^{1-x_1} * (x_1 - 1) * cos(x_2 * ln(B)) - A^{1-x_1} * x_2 * sin(x_2 * ln(A)) + A^{1-x_1} * (x_1 - 1) * cos(x_2 * ln(A))) * ((1 - x_1)^2 + x_2^2)^{-1} =$

$(B^{1-x_1} * x_2 * sin(x_2 * ln(B)) + B^{1-x_1} * (1 - x_1) * cos(x_2 * ln(B)) - A^{1-x_1} * x_2 * sin(x_2 * ln(A)) - A^{1-x_1} * (1 - x_1) * cos(x_2 * ln(A))) * ((1 - x_1)^2 + x_2^2)^{-1} =$

$(B^{1-x_1} * (x_2 * sin(x_2 * ln(B)) + (1 - x_1) * cos(x_2 * ln(B))) - A^{1-x_1} * (x_2 * sin(x_2 * ln(A)) - (1 - x_1) * cos(x_2 * ln(A)))) * ((1 - x_1)^2 + x_2^2)^{-1} =$

$(B^{1-x_1} * sin(x_2 * ln(B) + arctan((1 - x_1)/x_2)) - A^{1-x_1} * sin(x_2 * ln(A) + arctan((1 - x_1)/x_2))) * ((1 - x_1)^2 + x_2^2)^{-1/2}$